

Bayesian optimization with unknown constraints for the design and real-time control of stormwater systems

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ABSTRACT

Despite the prevalence of input-constrained optimization methods in the design and control of stormwater systems, these methods often do not align with the stated desires of designers. For example, it is often specified that flooding must not occur under real-time control (RTC). However, this constraint is defined in the output space and thereby has *a priori* unknown relationship to the controllable inputs. Bayesian Optimization with Unknown Constraints (BOUC) accurately approximates these constraints defined in the output space. We demonstrate the utility of BOUC in the structural and operational design of five drainage networks from the real-time control benchmarking library *pystorms*. We also address four limitations in the original version of *pystorms*. (1) The cost function for some scenarios is not strongly influenced by the actuators provided. (2) Other scenarios have cost functions lenient enough that local rule-based controllers are sufficient to score perfectly. (3) Sensor and actuator faults and noise are not represented, though they are a key challenge of practical RTC implementation. (4) The benchmark controllers necessary to avoid an inappropriate comparison of the performance of new control algorithms with the passive system are not included. We present revised scenarios improving feasibility (1) and increasing discriminative power (2) while implementing randomized faults (3). Heuristic local and coordinated controllers are tuned using BOUC to benchmark future control algorithms (4). This study introduces an optimization method well-suited to stormwater design and operation while also expanding an existing benchmarking tool to support further development and fair comparison of methods.

1. Comparison with other optimization methods

Convergence rates are compared with (1) Bayesian Optimization using Expected Improvement acquisition function, (2) Dual Annealing, and (3) Differential Evolution. BOUC generally achieves better scores given similar numbers of function evaluations. However, the model update step is expensive compared to the other optimizations. Thus, BOUC's sample efficiency is more helpful when the objective and constraint functions are expensive to evaluate. Results are presented for two of the *pystorms2.0* scenarios and a benchmark problem.

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Bayesian optimization with unknown constraints for stormwater

Method	Normalized Final Cost	Time Elapsed	Number of model runs
Bayesian Optimization with Unknown Constraints	1.0	8 minutes	75
Bayesian Optimization (Expected Improvement)	1.06	4.3 minutes	75
Dual Annealing	1.03	46 seconds	75
Differential Evolution	1.012	42 seconds	100*

Table 1

Optimizing the constant flow controller (2 decision variables) for scenario *theta*. *The library used did not allow direct specification of number of function evaluations.

1.1. *Theta*

36

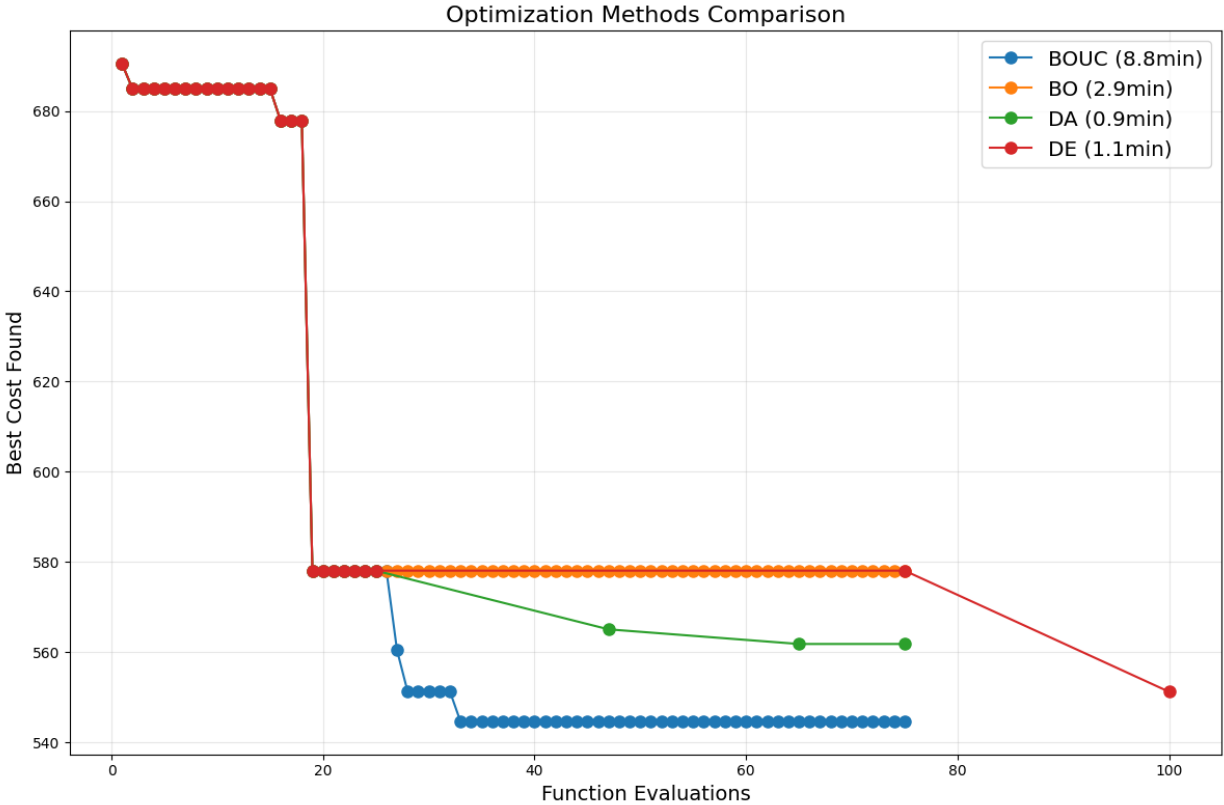


Figure 1: Cost against number of model runs for *theta*. Costs for the first evaluations track together as all models are given the same randomized initial points.

Bayesian optimization with unknown constraints for stormwater

Method	Normalized Final Cost	Time Elapsed	Number of model runs
Bayesian Optimization with Unknown Constraints	1.0	5.4 minutes	30
Bayesian Optimization (Expected Improvement)	1.14	4.4 minutes	30
Dual Annealing	1.23	3.6 minutes	35
Differential Evolution	1.30	1.8 minutes	25

Table 2

Optimizing the constant flow controller (9 decision variables) for scenario *gamma*.

37

1.2. *Gamma*

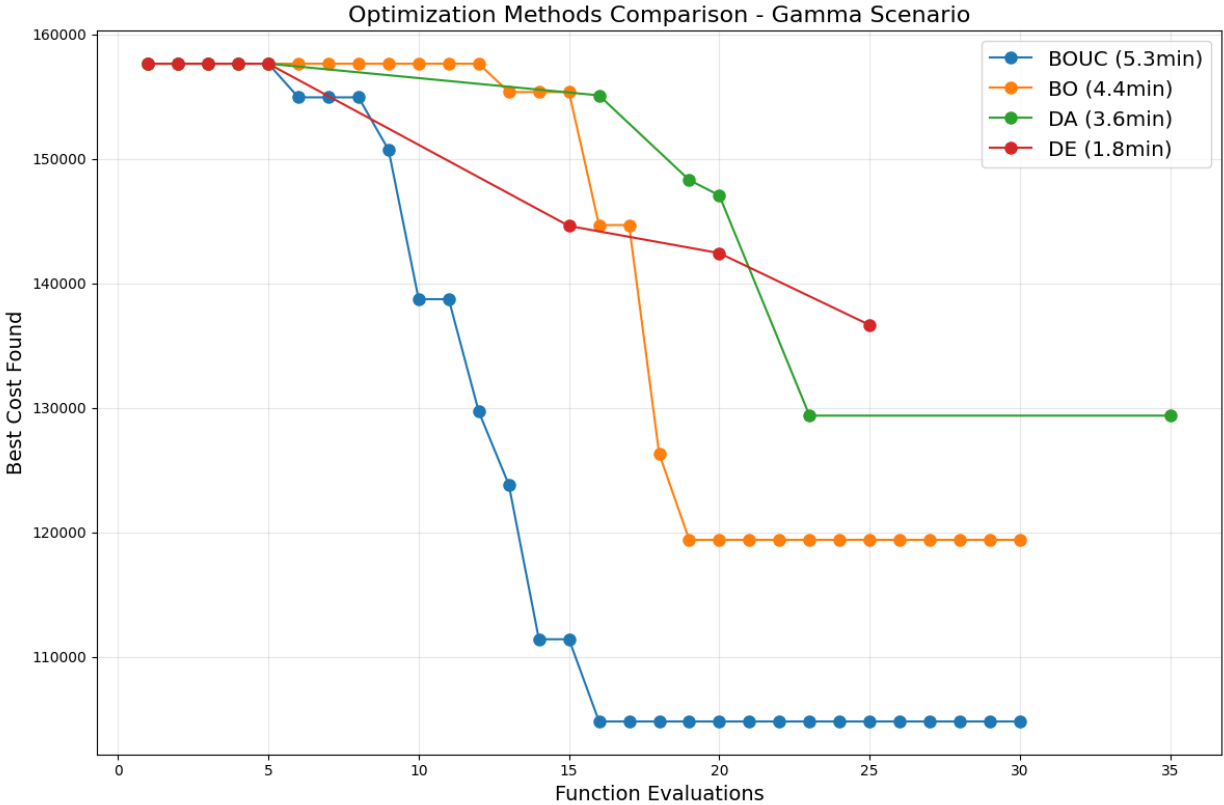


Figure 2: Cost against number of model runs for *gamma*. Costs for the first evaluations track together as all models are given the same randomized initial points.

Method	Normalized Final Cost	Time Elapsed	Function evaluations
Bayesian Optimization with Unknown Constraints	1.0	3.7 minutes	75
Bayesian Optimization (Expected Improvement)	3.81	1.56 minutes	75
Dual Annealing	2.42	< 1 second	75
Differential Evolution	3.49	< 1 second	100

Table 3

Minimizing the Ackley function with two disconnected feasible regions. This function has many local minima which make it a challenging optimization problem.

1.3. Ackley Function with discontinuous feasible region.

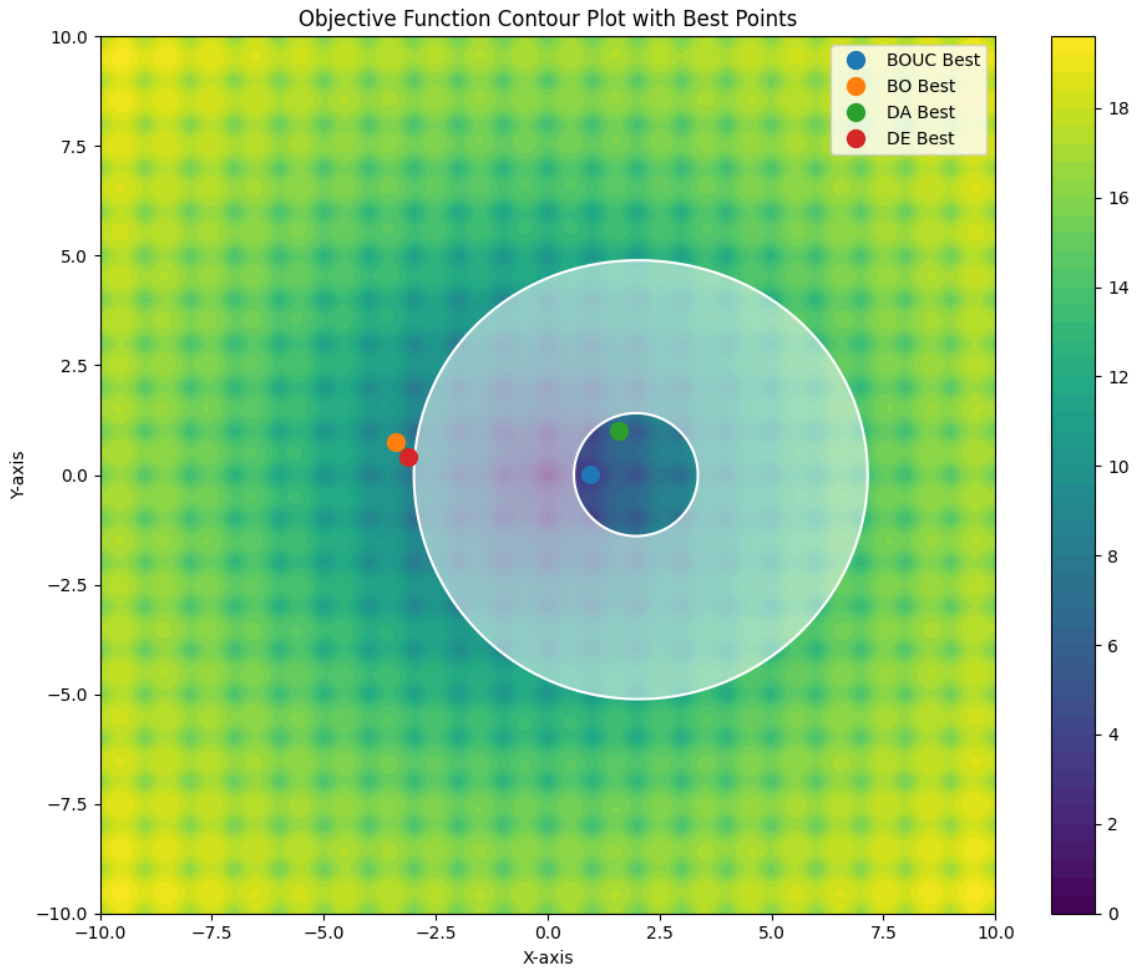


Figure 3: Two-dimensional Ackley function with infeasible region shaded in white. The best result for each model is shown with BOUC approaching the feasible global minimum and the other three methods near local minima.

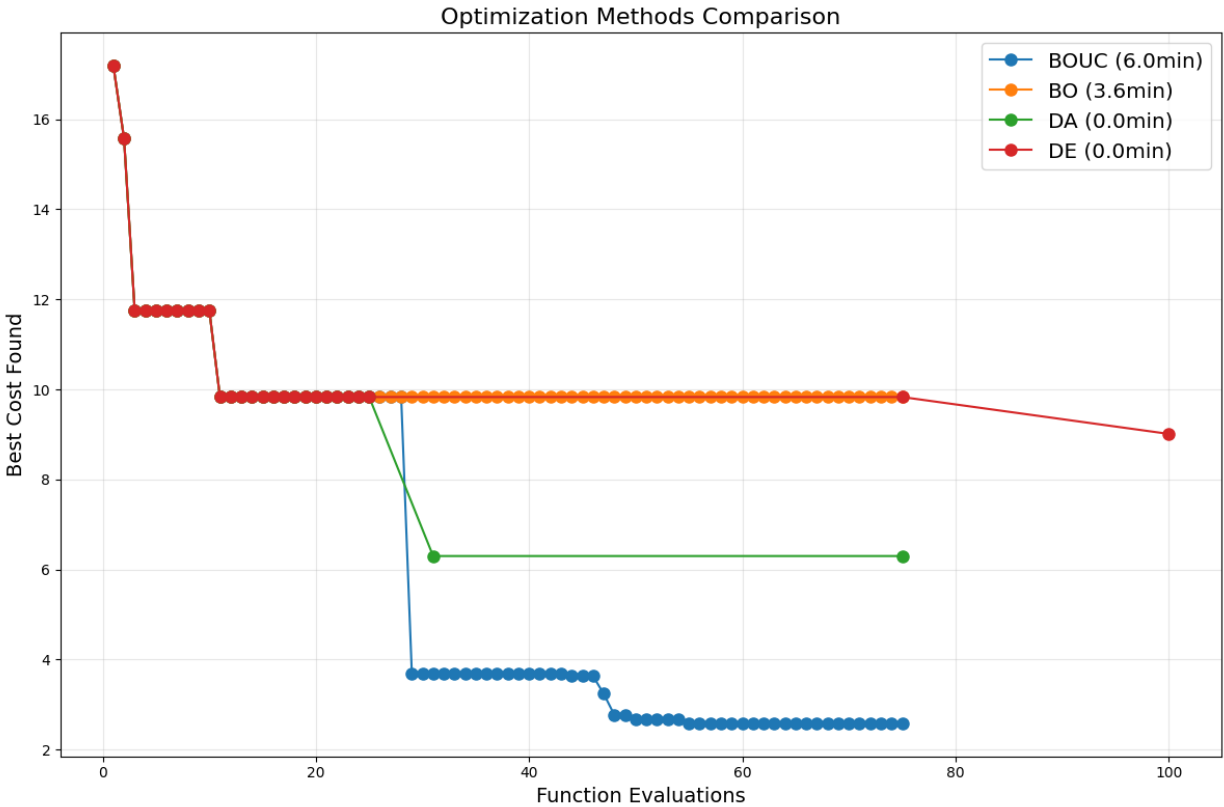


Figure 4: Cost against number of function evaluations for Ackley. Costs for the first evaluations track together as all models are given the same randomized initial points.

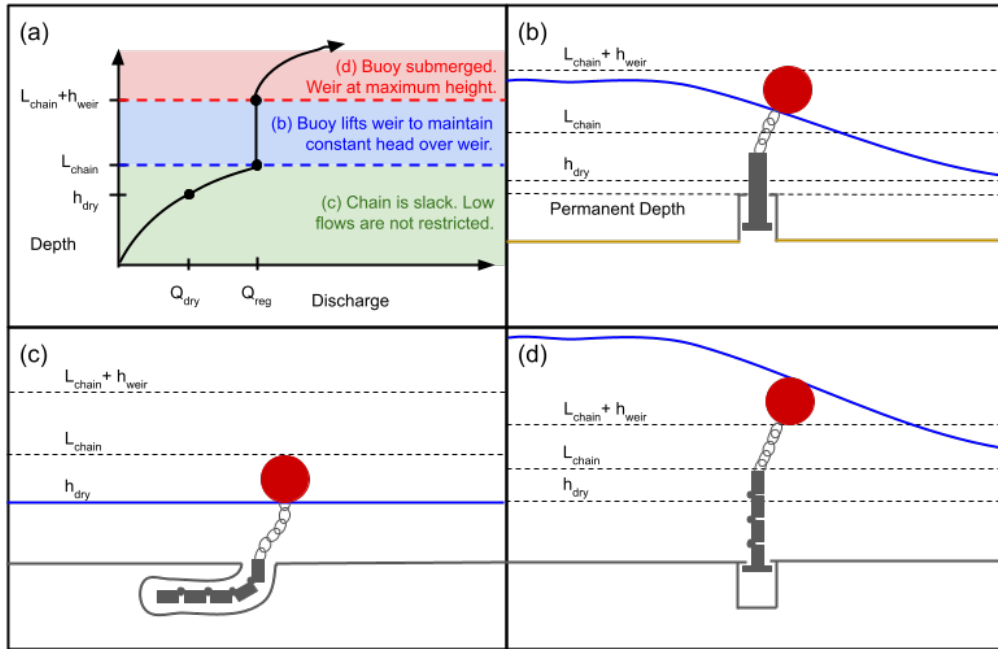


Figure 5: A mechanically dynamic weir requiring basic manufacturing and assembly capabilities. **(a)** A conceptual illustration of the depth-discharge relation where h_{dry} is the dry-weather or baseflow head over the weir, L_{chain} is the length of the chain(s) connecting the weir to the buoy(s) which determines when flow regulation starts, and h_{weir} is the height of the weir not including the chain which determines the range of depths for which flow will be regulated. Q_{dry} is the flow corresponding to h_{dry} while Q_{reg} is the desired constant flow rate. **(b)** The simplest implementation uses a solid weir stored vertically and attached to buoys with an adjustable-length chain (L_{chain}). The solid line is the actual water level while dashed lines correspond to the depth-discharge relation in (a). This design may be appropriate for the level control structure of a small lake or CSO regulator (e.g., *alpha* or *delta*). Note that this structure has permanent storage. That permanent storage could be eliminated by burying the weir further down. This sketch provides an example of the region in the depth-discharge relation where flows are kept constant. **(c)** For reduced excavation depth the weir could be stored horizontally as a hinged wall which is rigid against the flow direction but may fold in the other direction (e.g., *epsilon*). As the chain is slack and cannot raise the weir, low flows are not restricted. **(d)** The weir may be stored in a spiral if the hinges are close together (also *epsilon*). High flows beyond a certain head are not restricted as the weir is fully lifted and the buoy is submerged. Flows are still less than they would be without the weir present as it blocks cross-sectional area. Watertightness of the hinges is not necessary for effective flow regulation.

2. A mechanically dynamical weir without need for external actuation

A conceptual design for a mechanically dynamic weir is shown in SI Figure 5. Buoys are attached by adjustable chains to the top of a weir wall. That wall may either be hinged so as to store horizontally under the conduit or be straight and descend into a pit. The former is likely more applicable to underground conveyances such as sewer pipes while the latter could regulate outflows from a retention pond.

3. Performance of benchmark controllers under varying noise and faults

A suite of plots is provided for all five scenarios revised in this work. These are organized in the manner of the main text as *theta*, *alpha*, *gamma*, *delta*, and *epsilon*. Within each scenario are figures showing:

- Comparison of varying control approaches under each noise level.
- Comparison of each control approach under varying noise levels.

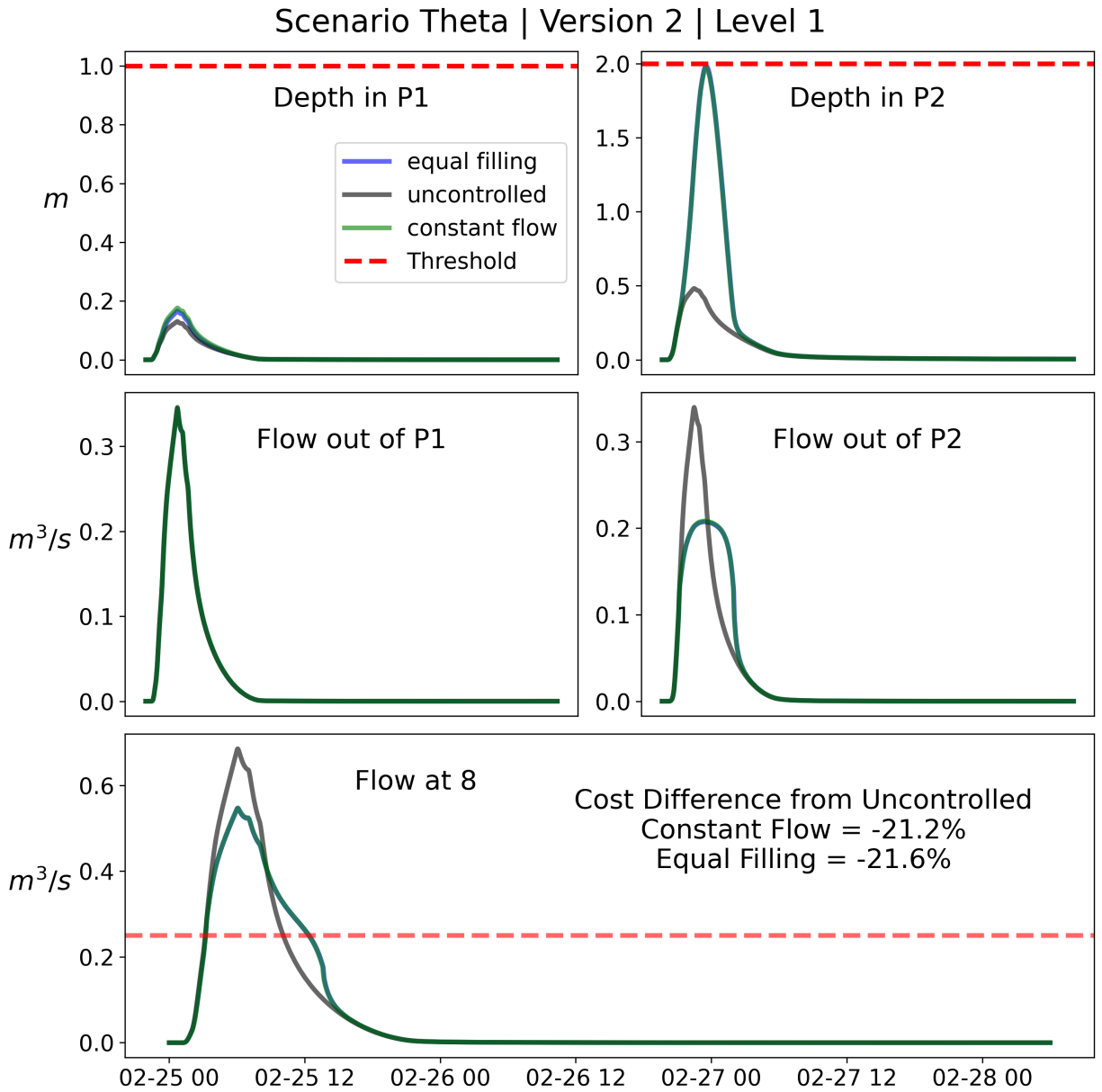


Figure 6: Theta noise level 1. P1 and P2 are ponds which drain to a common conduit called 8.

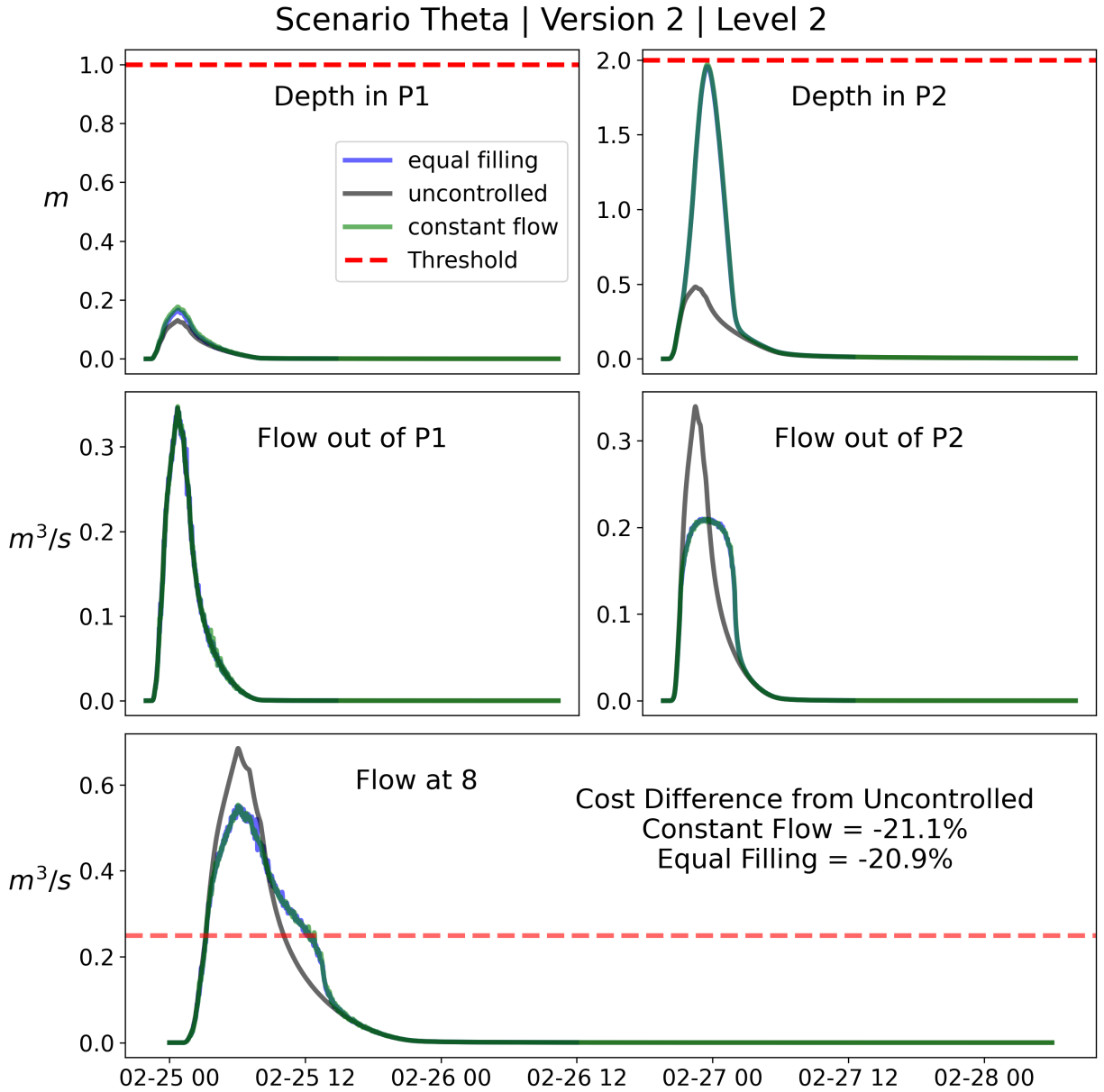


Figure 7: Theta noise level 2. P1 and P2 are ponds which drain to a common conduit called 8.

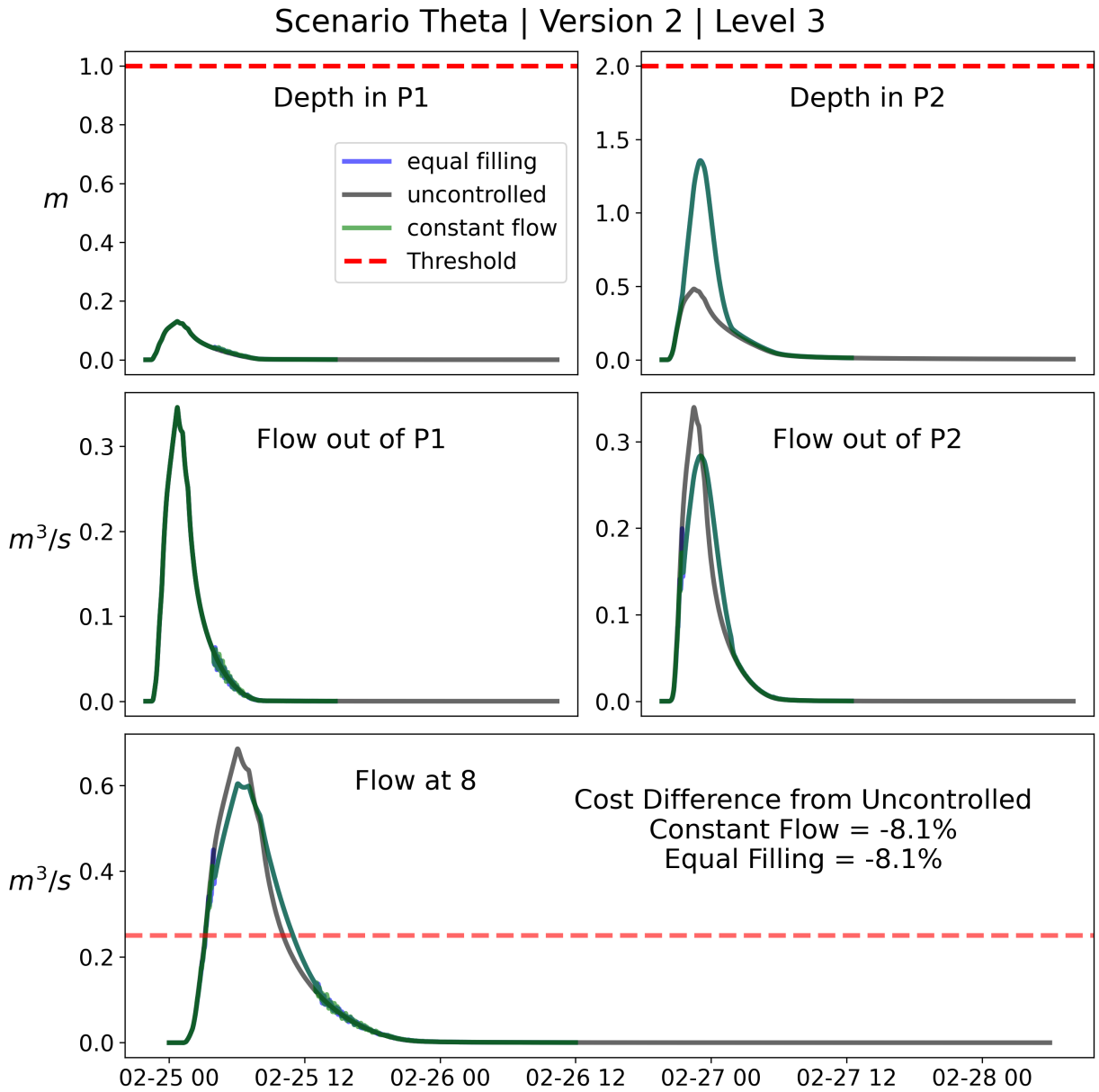


Figure 8: Theta noise level 3. P1 and P2 are ponds which drain to a common conduit called 8.

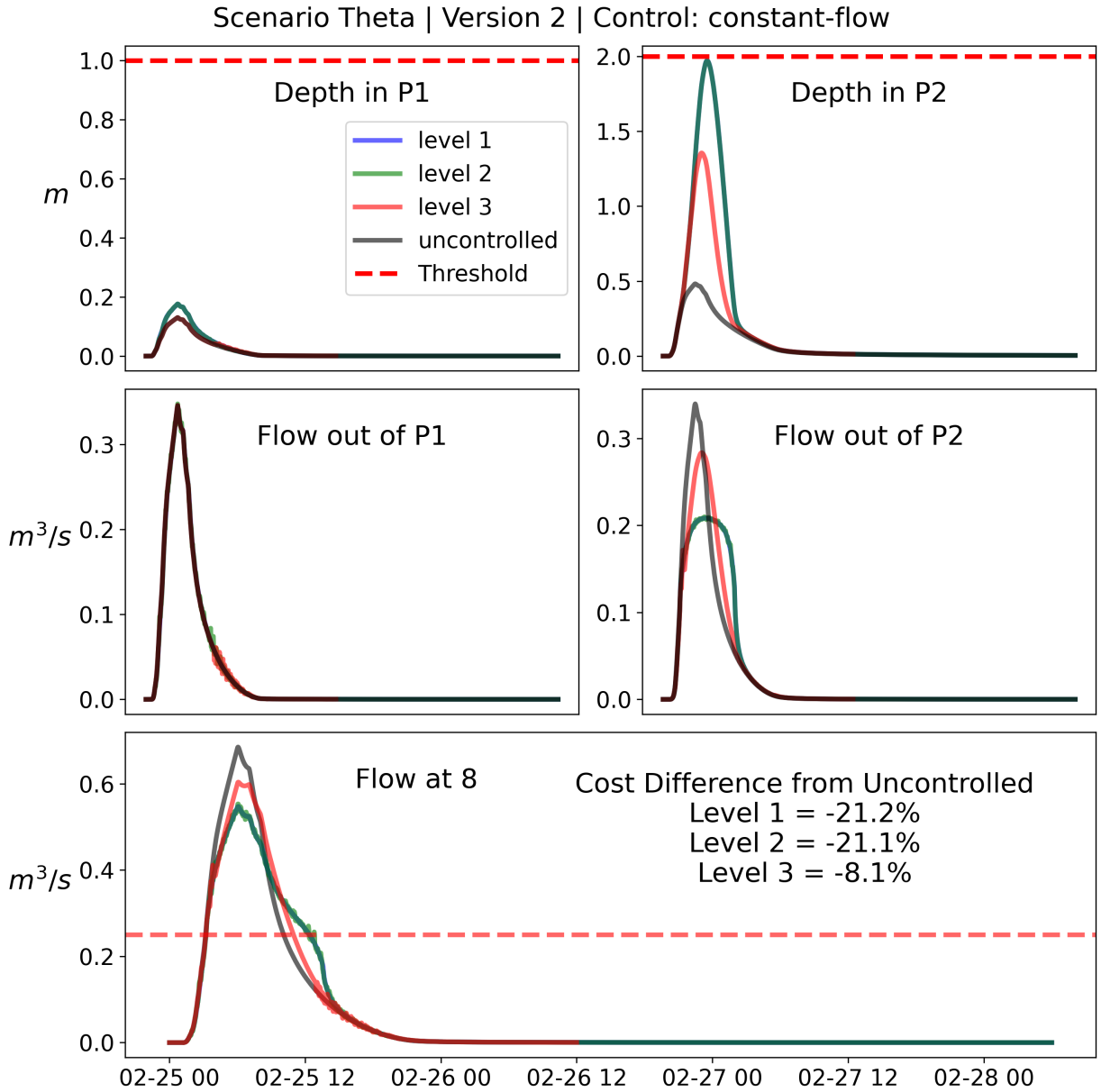


Figure 9: Theta constant flow controller. P1 and P2 are ponds which drain to a common conduit called 8.

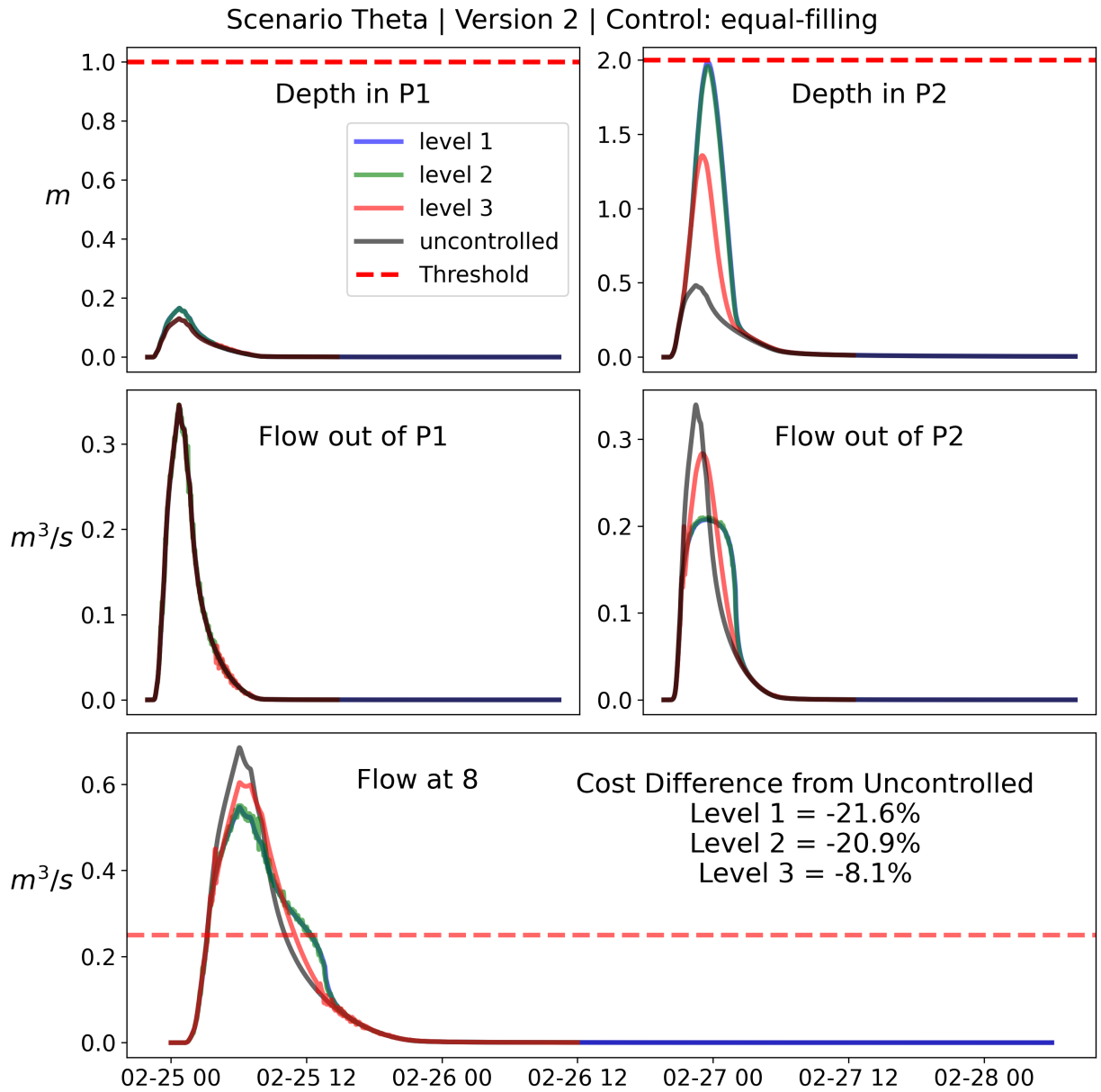


Figure 10: Theta equal filling controller. P1 and P2 are ponds which drain to a common conduit called 8.

Bayesian optimization with unknown constraints for stormwater

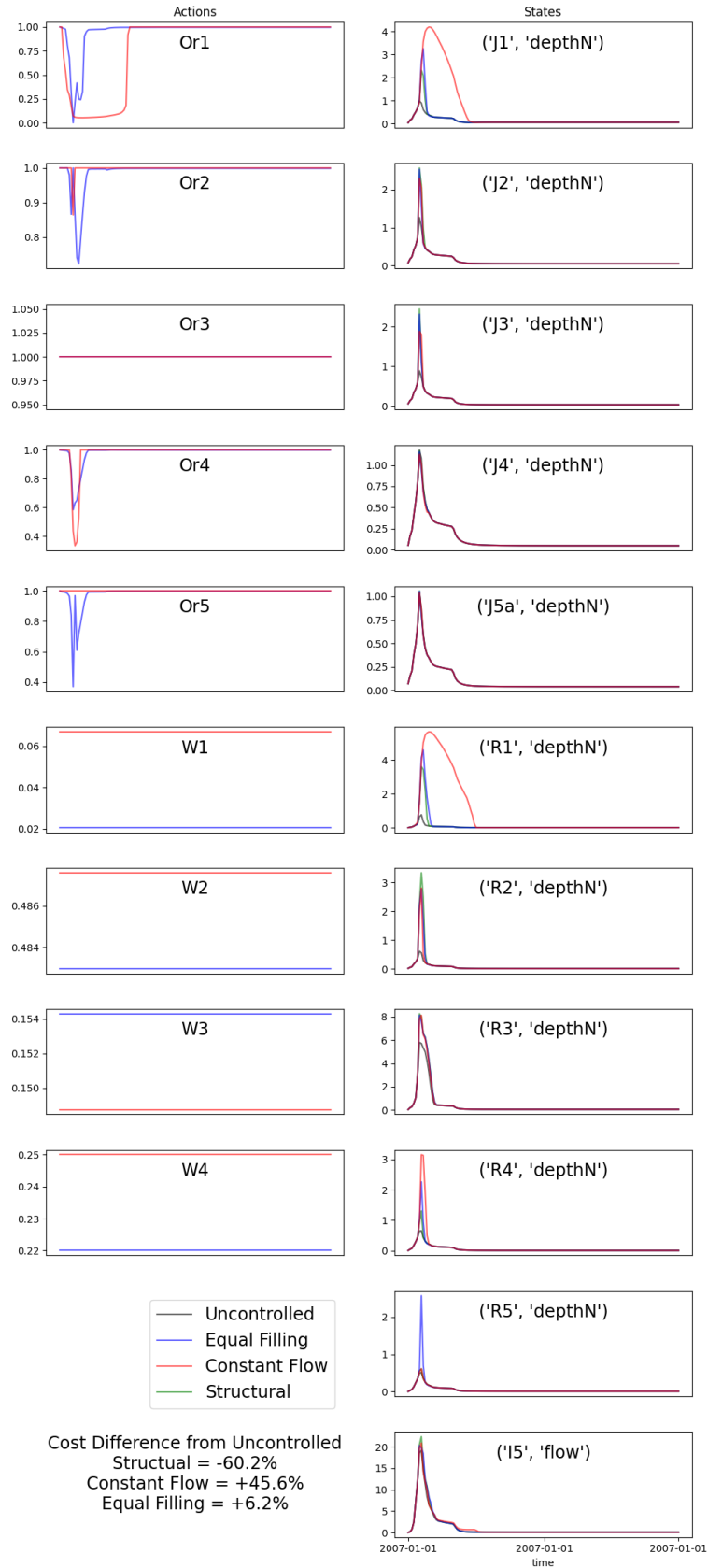


Figure 11: Alpha noise level 1

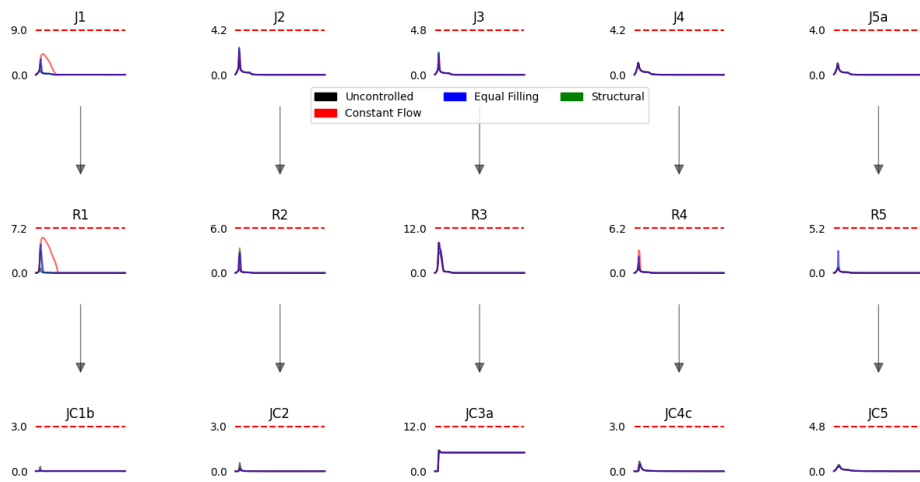


Figure 12: Alpha noise level 1

Bayesian optimization with unknown constraints for stormwater

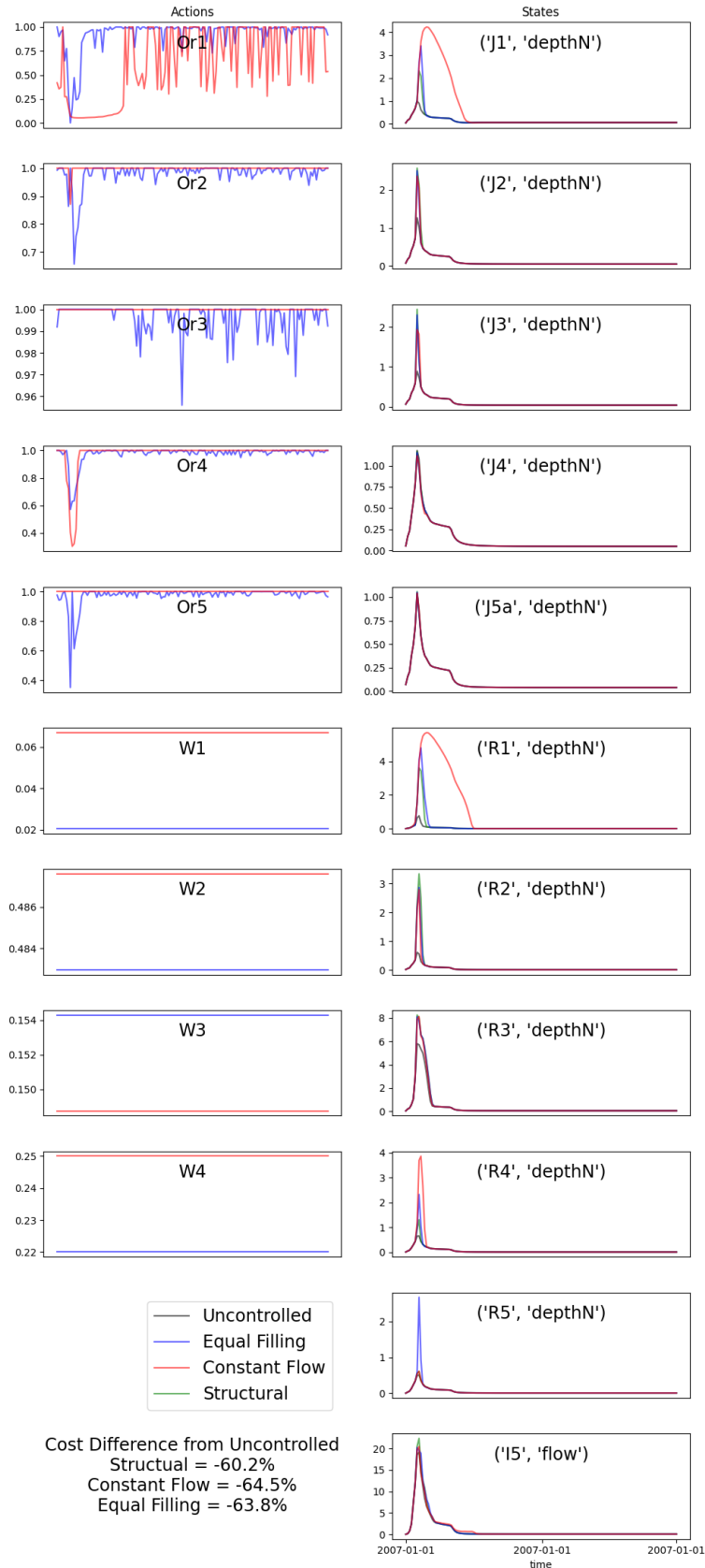


Figure 13: Alpha noise level 2

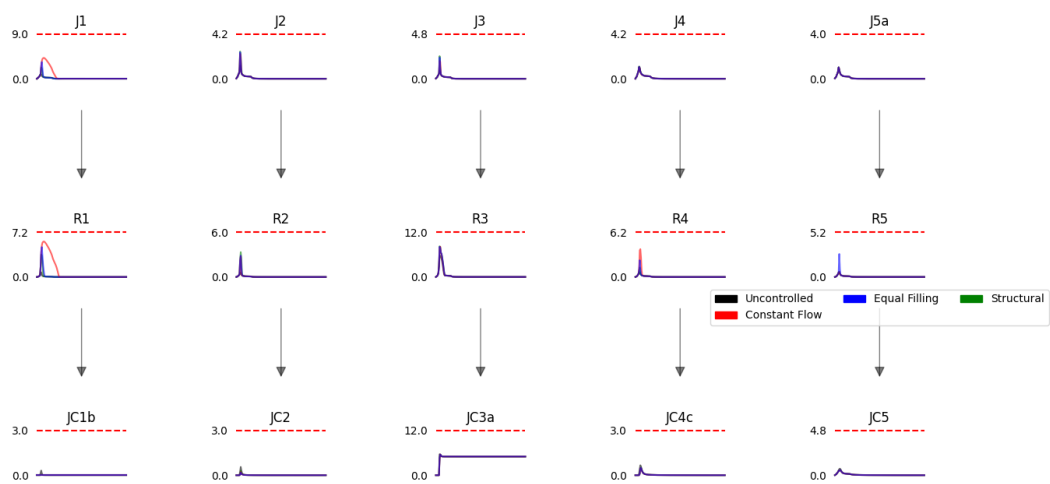


Figure 14: Alpha noise level 2

Bayesian optimization with unknown constraints for stormwater

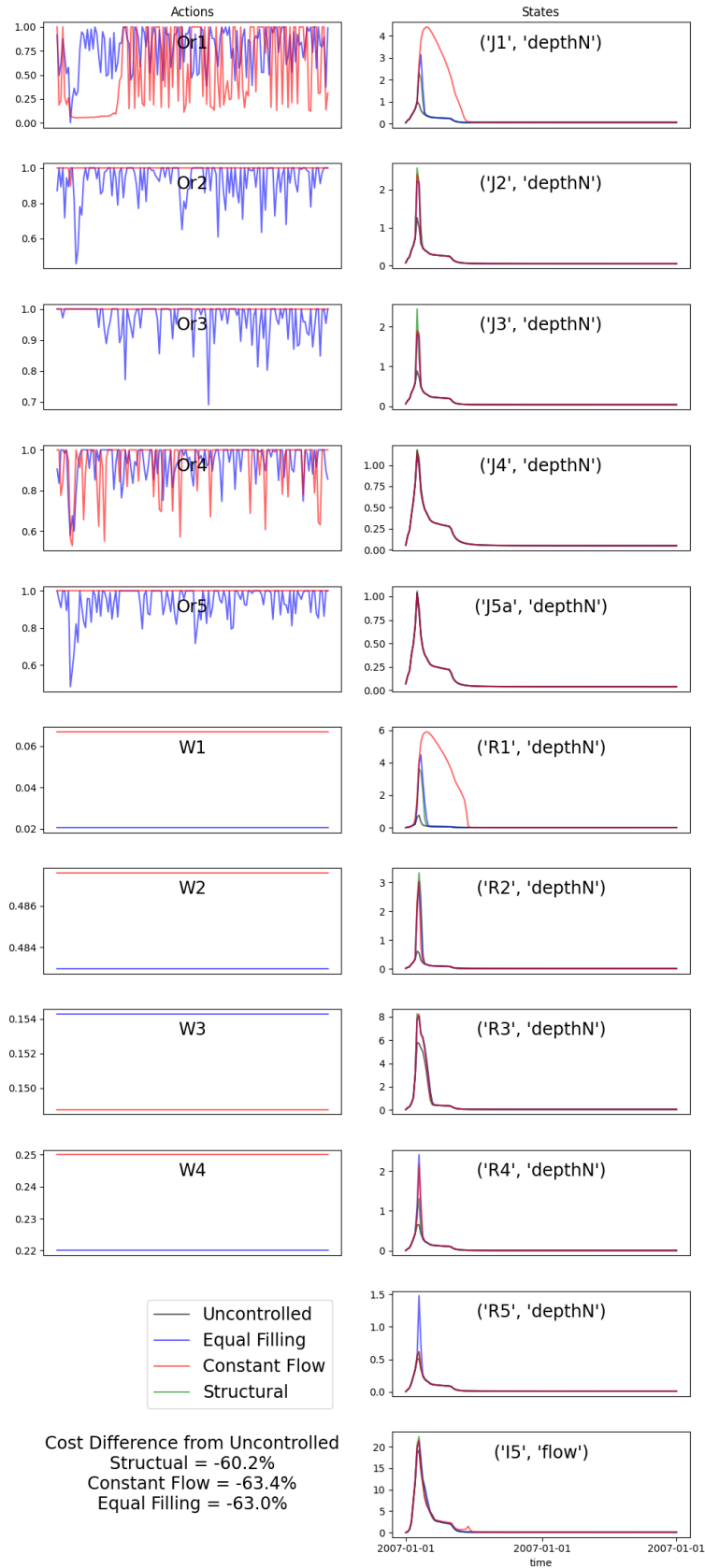


Figure 15: Alpha noise level 3

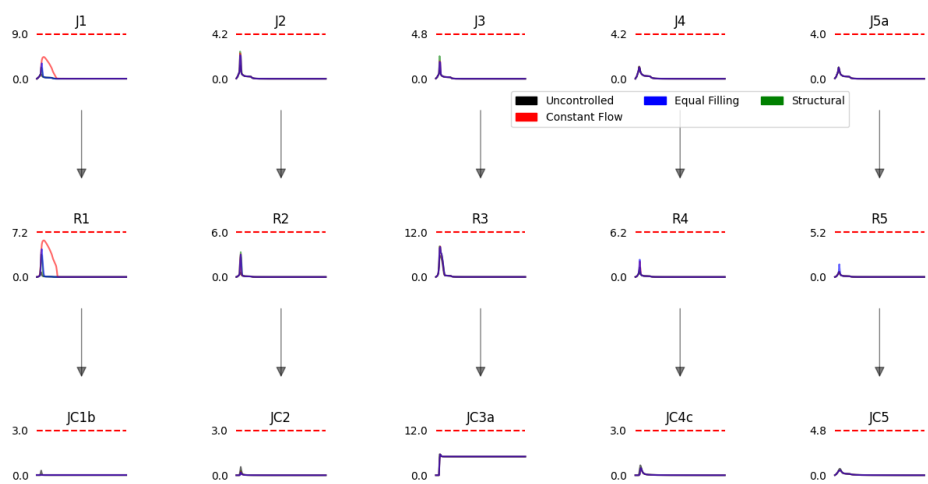


Figure 16: Alpha noise level 3

Bayesian optimization with unknown constraints for stormwater

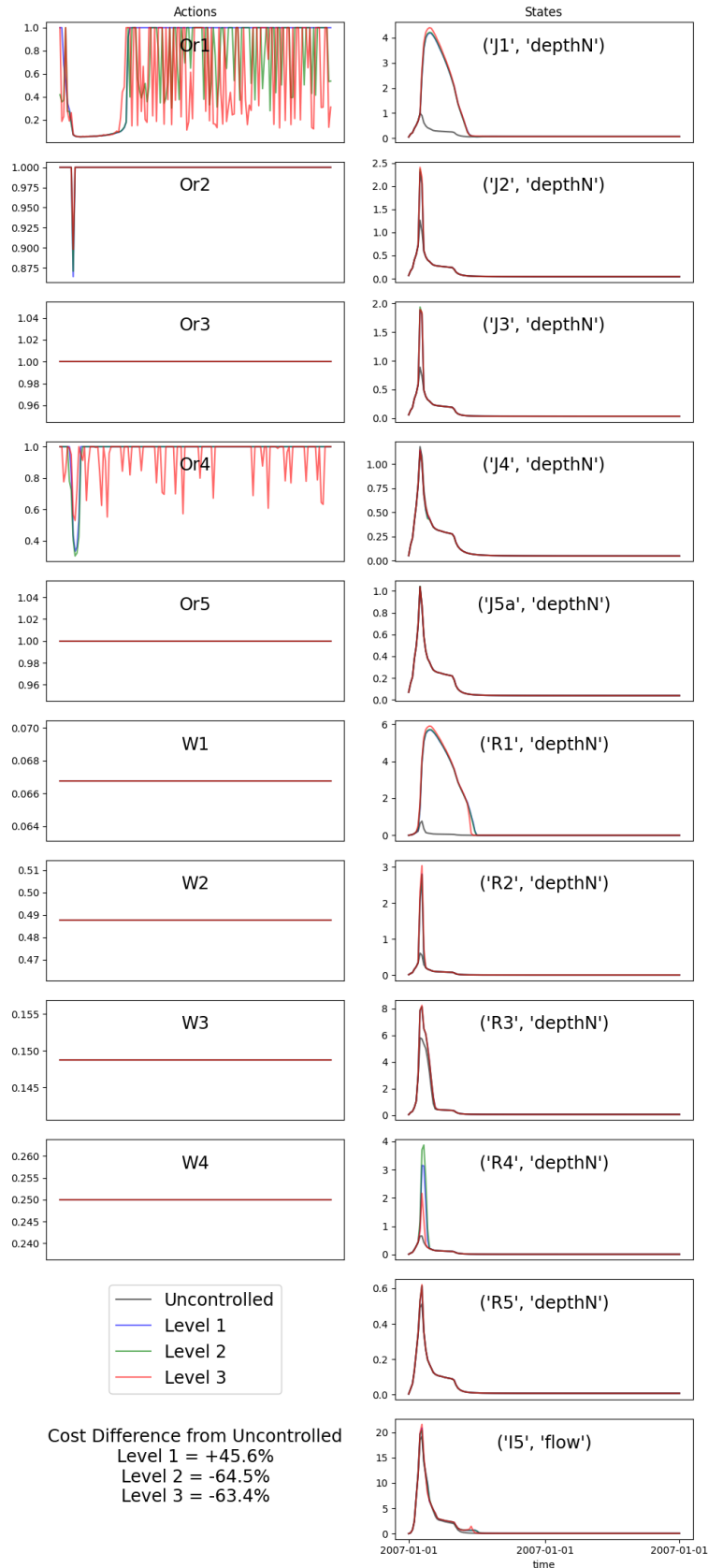


Figure 17: Alpha constant flow

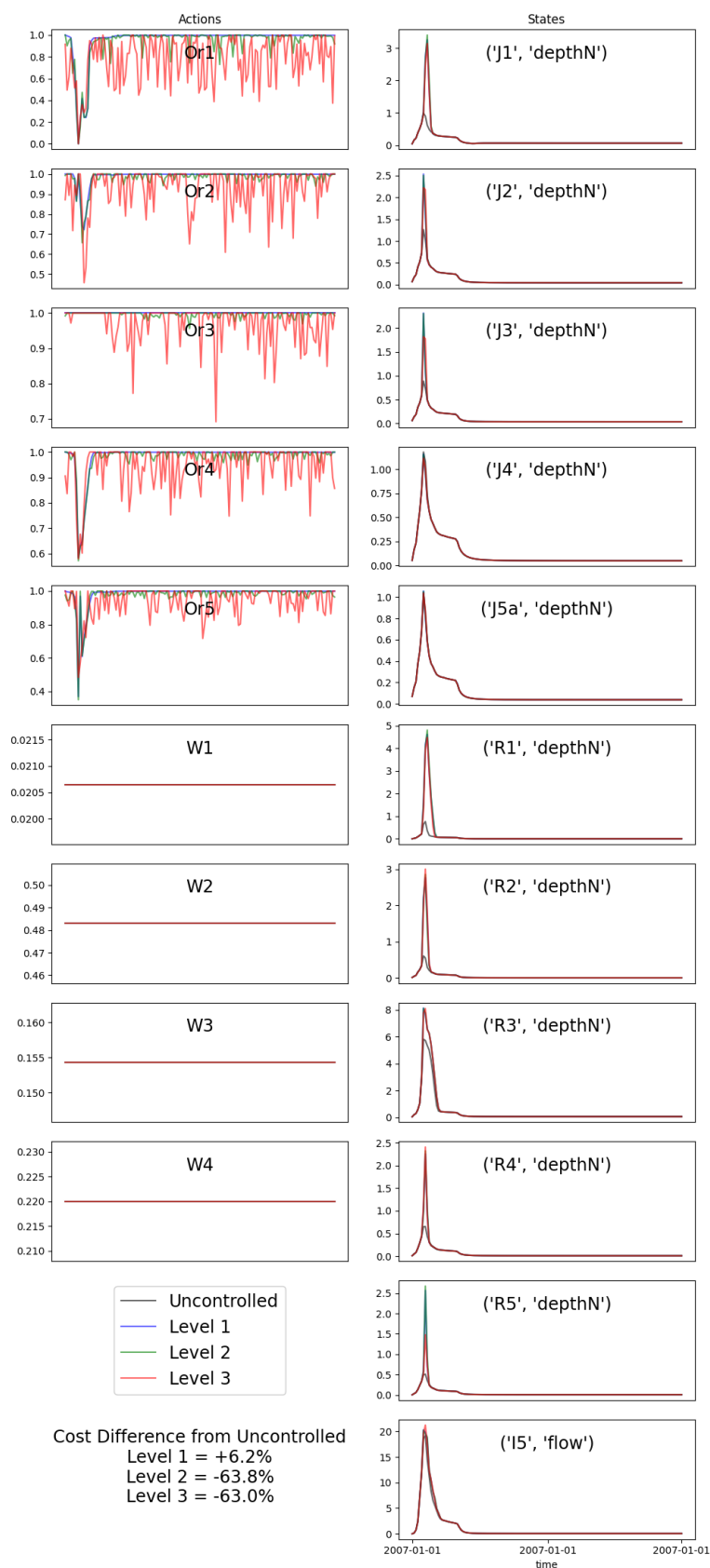


Figure 18: Alpha equal-filling-degree

Bayesian optimization with unknown constraints for stormwater

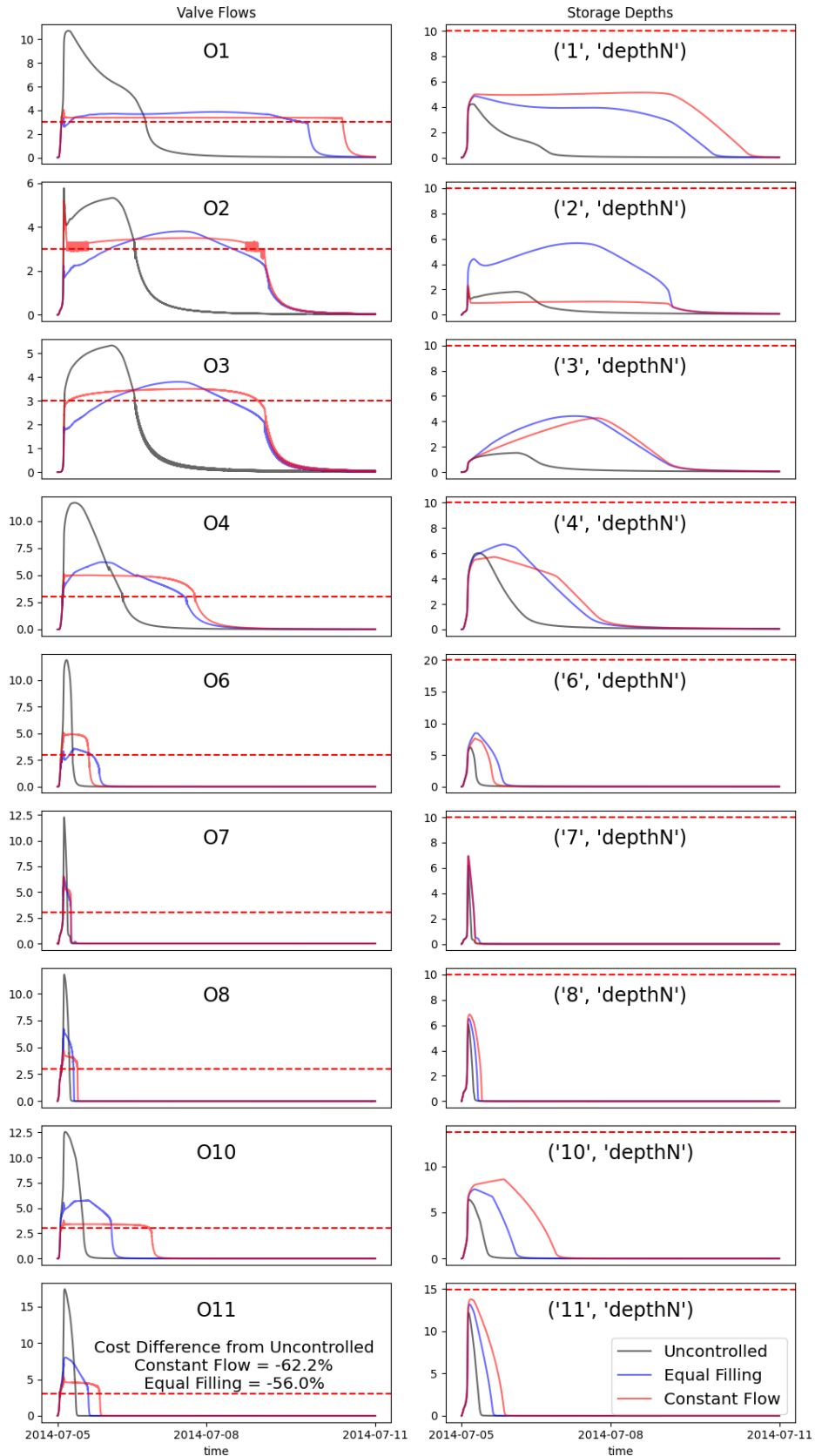


Figure 19: Gamma noise level 1

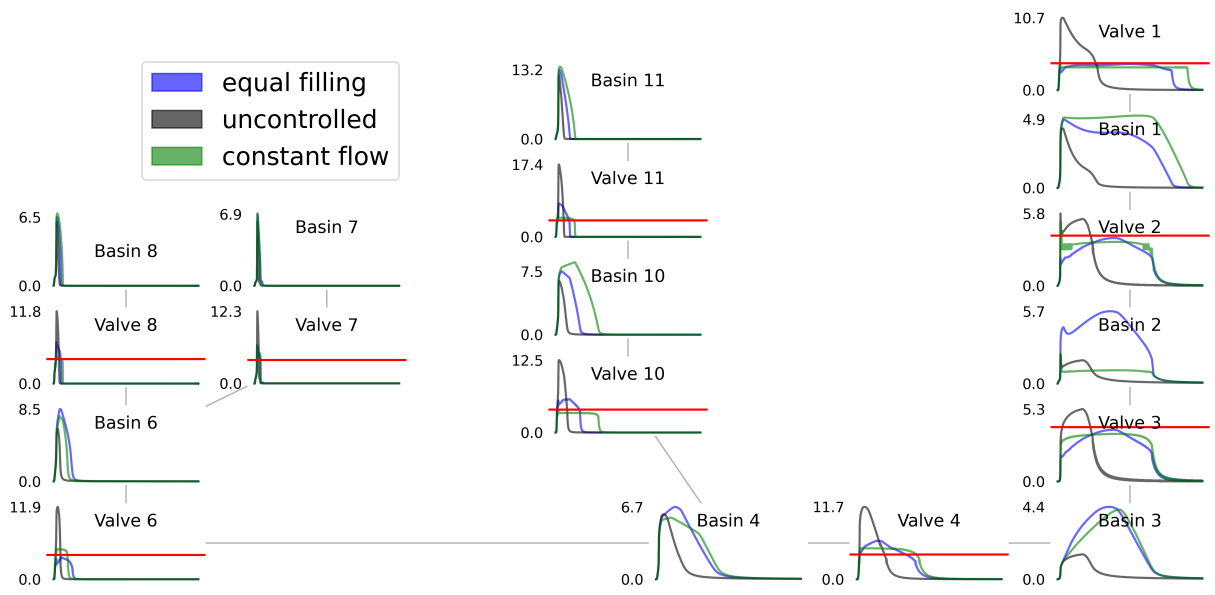


Figure 20: Gamma noise level 1

Bayesian optimization with unknown constraints for stormwater

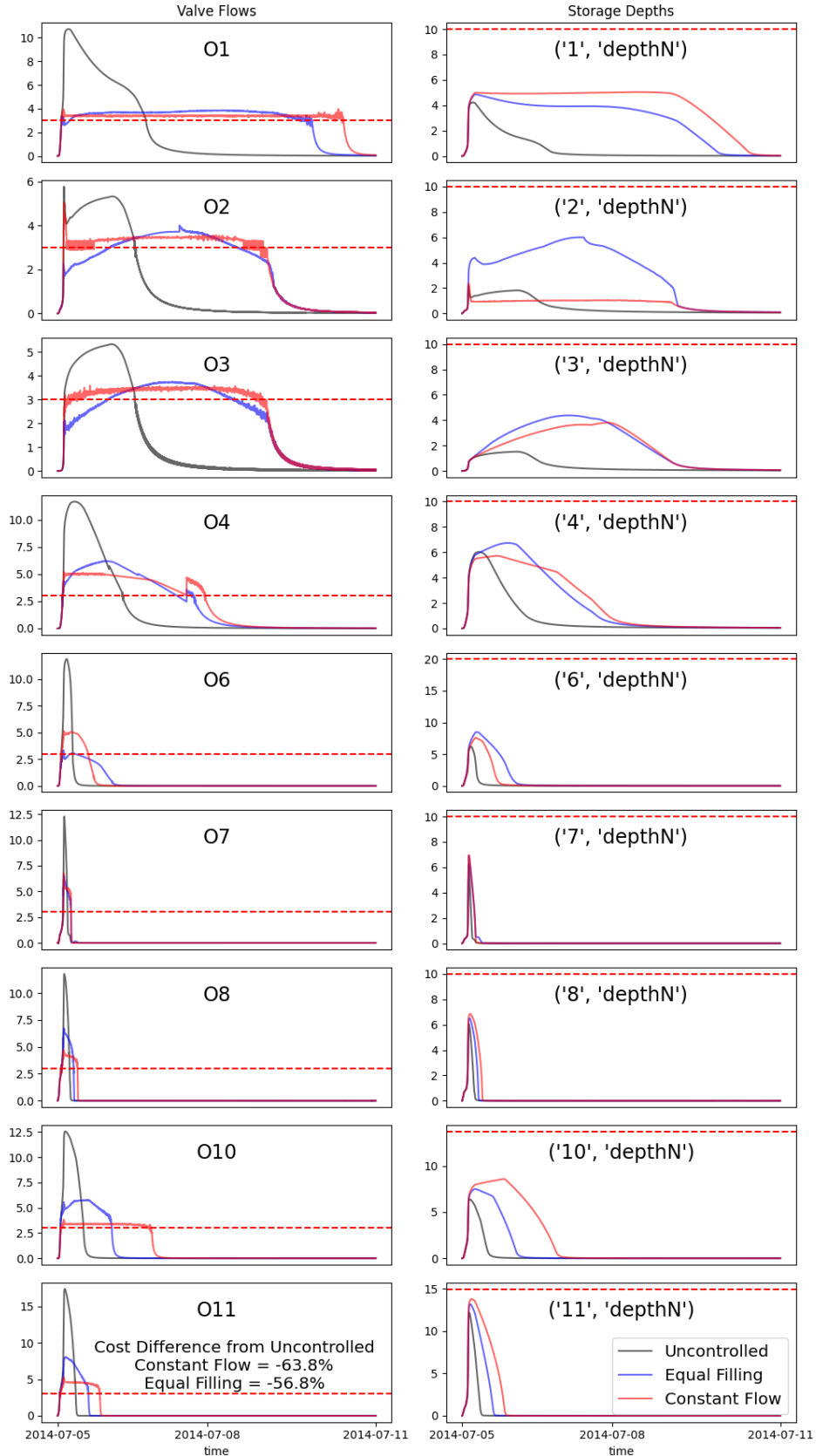


Figure 21: Gamma noise level 2

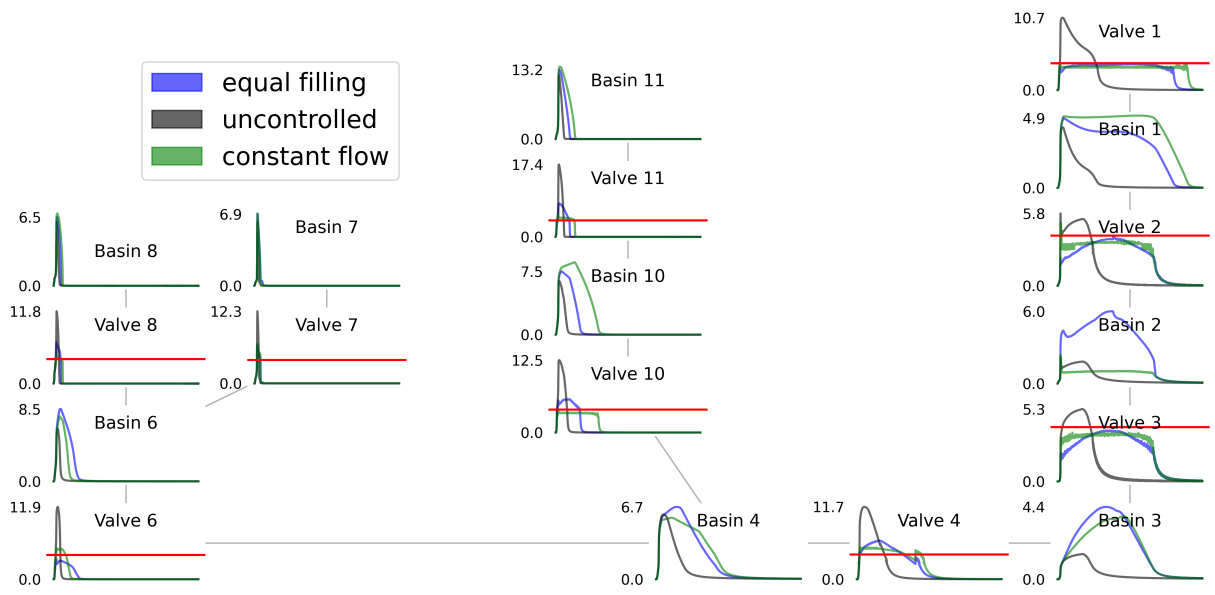


Figure 22: Gamma noise level 2

Bayesian optimization with unknown constraints for stormwater

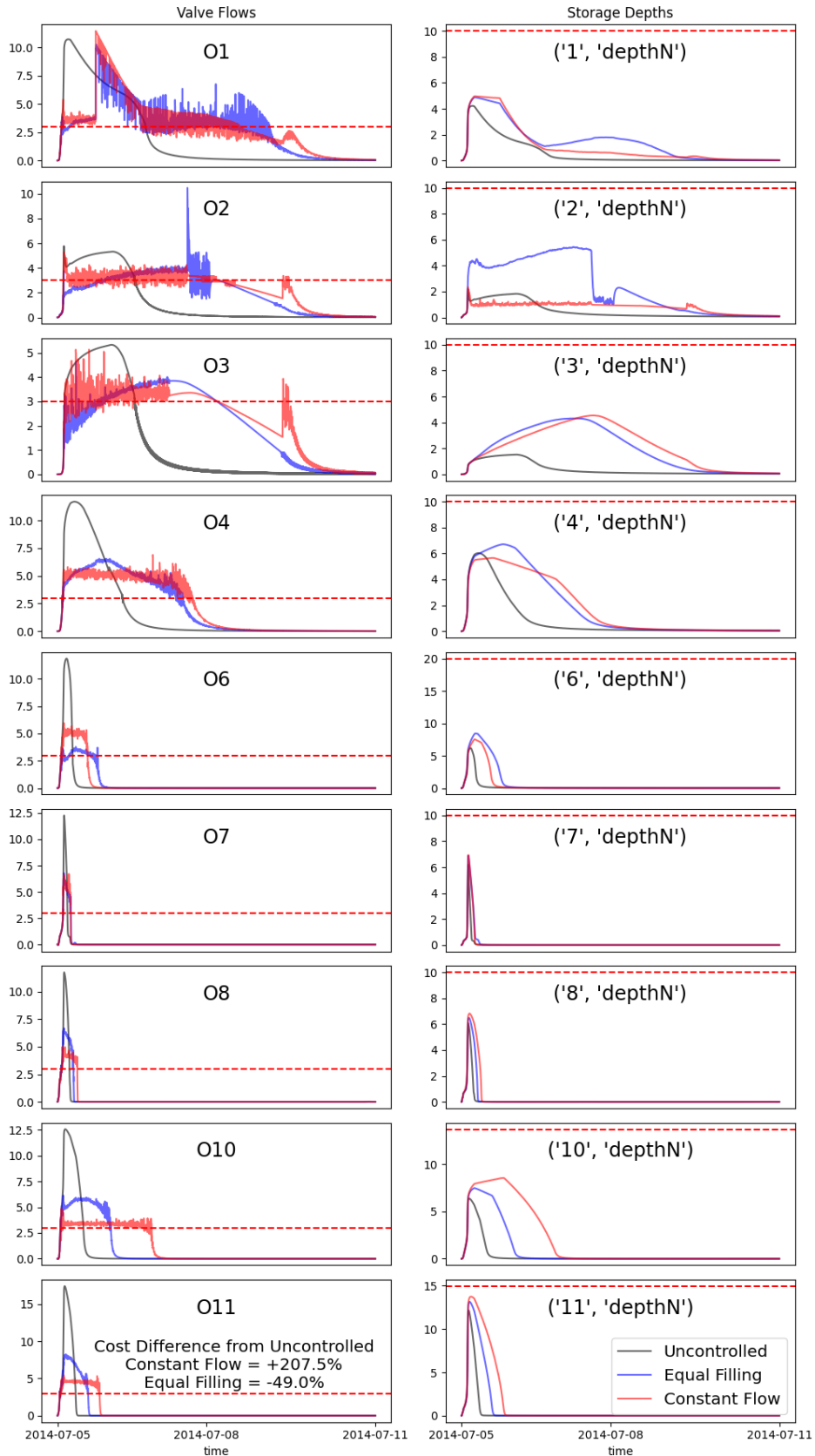


Figure 23: Gamma noise level 3

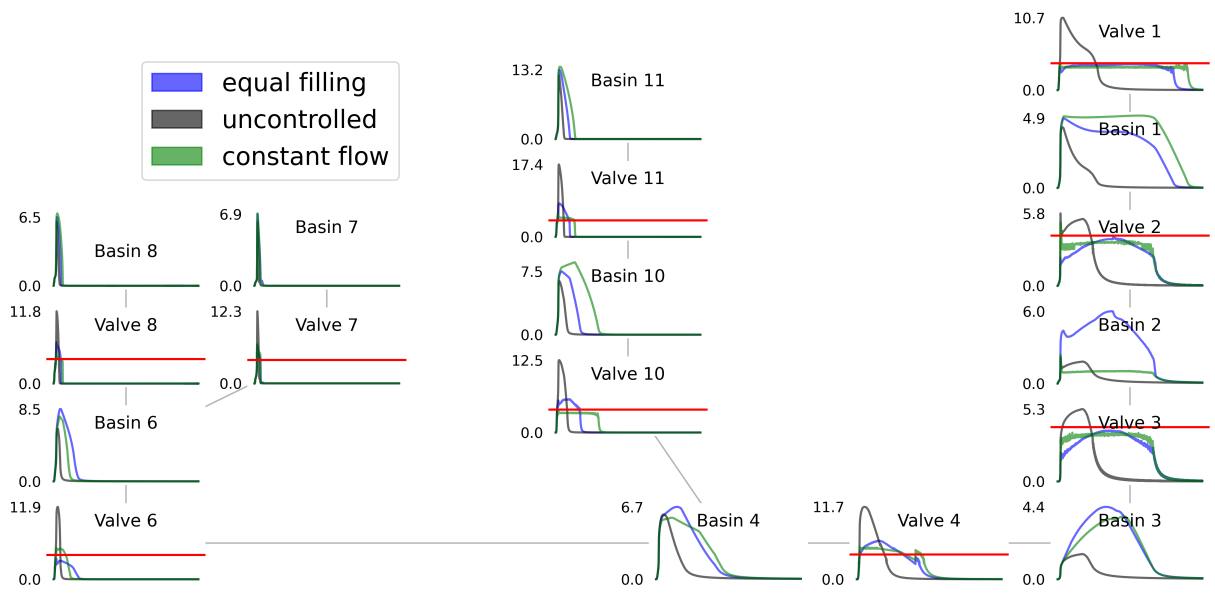


Figure 24: Gamma noise level 3

Bayesian optimization with unknown constraints for stormwater

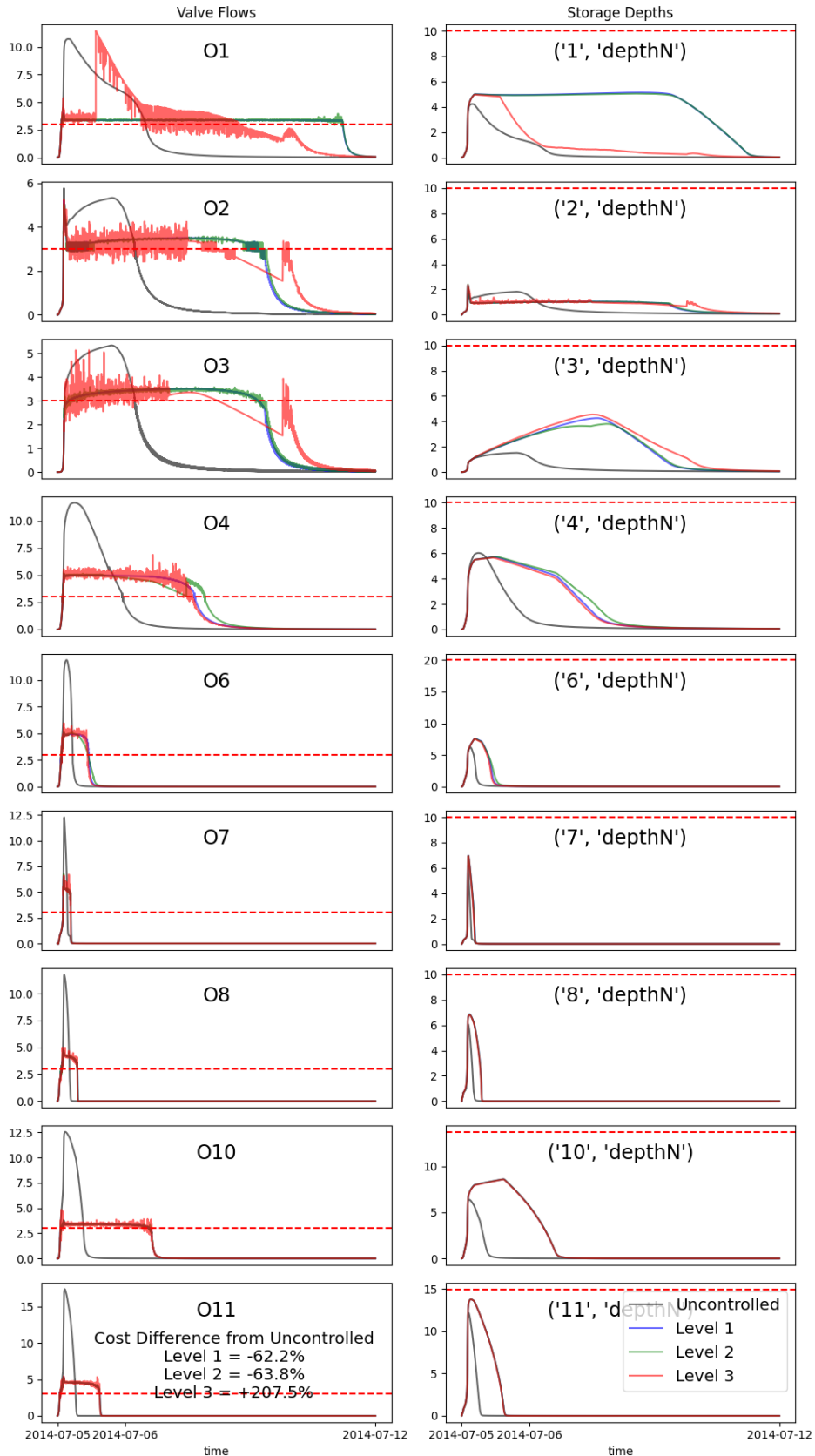


Figure 25: Gamma constant flow

Bayesian optimization with unknown constraints for stormwater

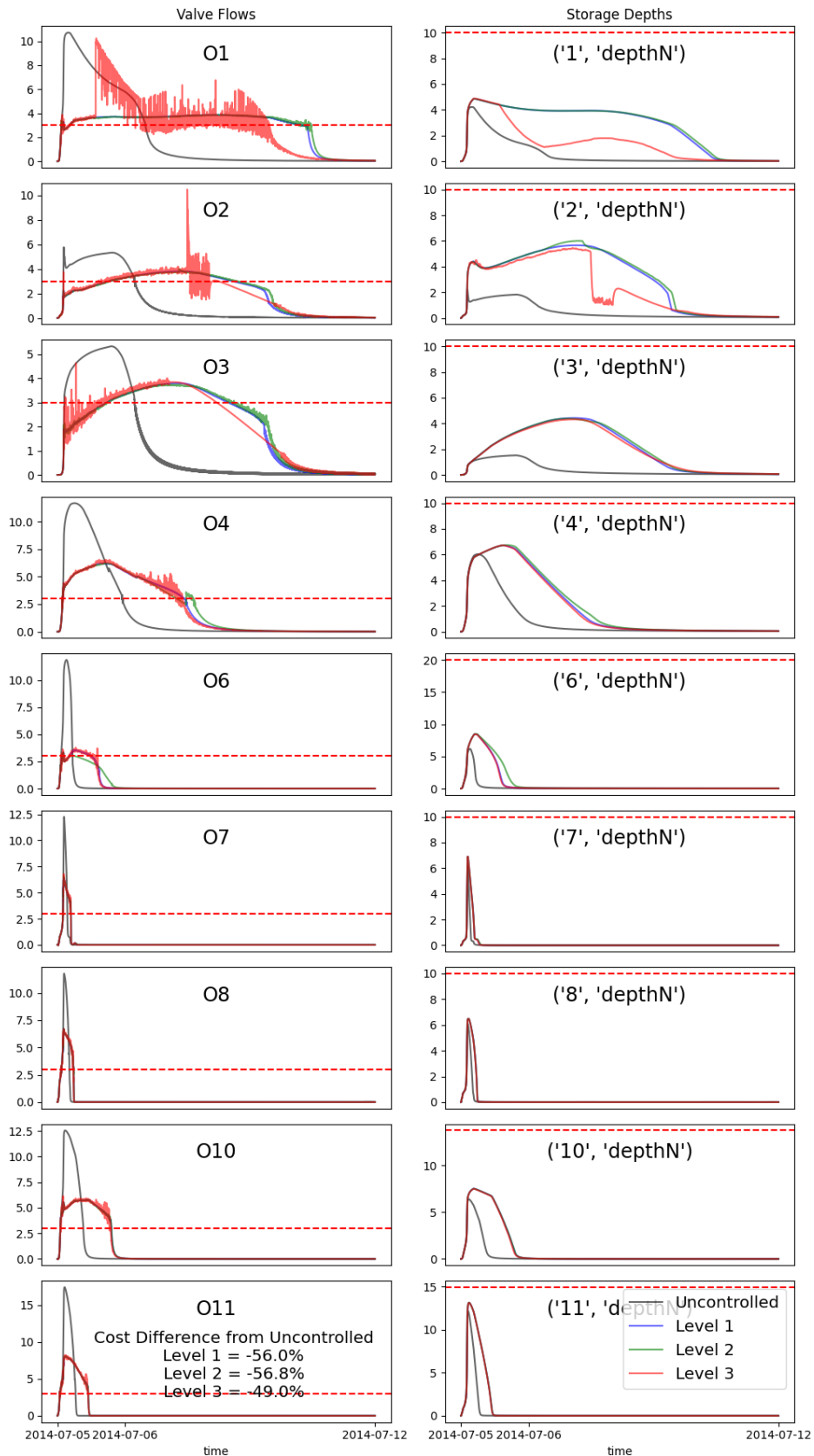


Figure 26: Gamma equal filling

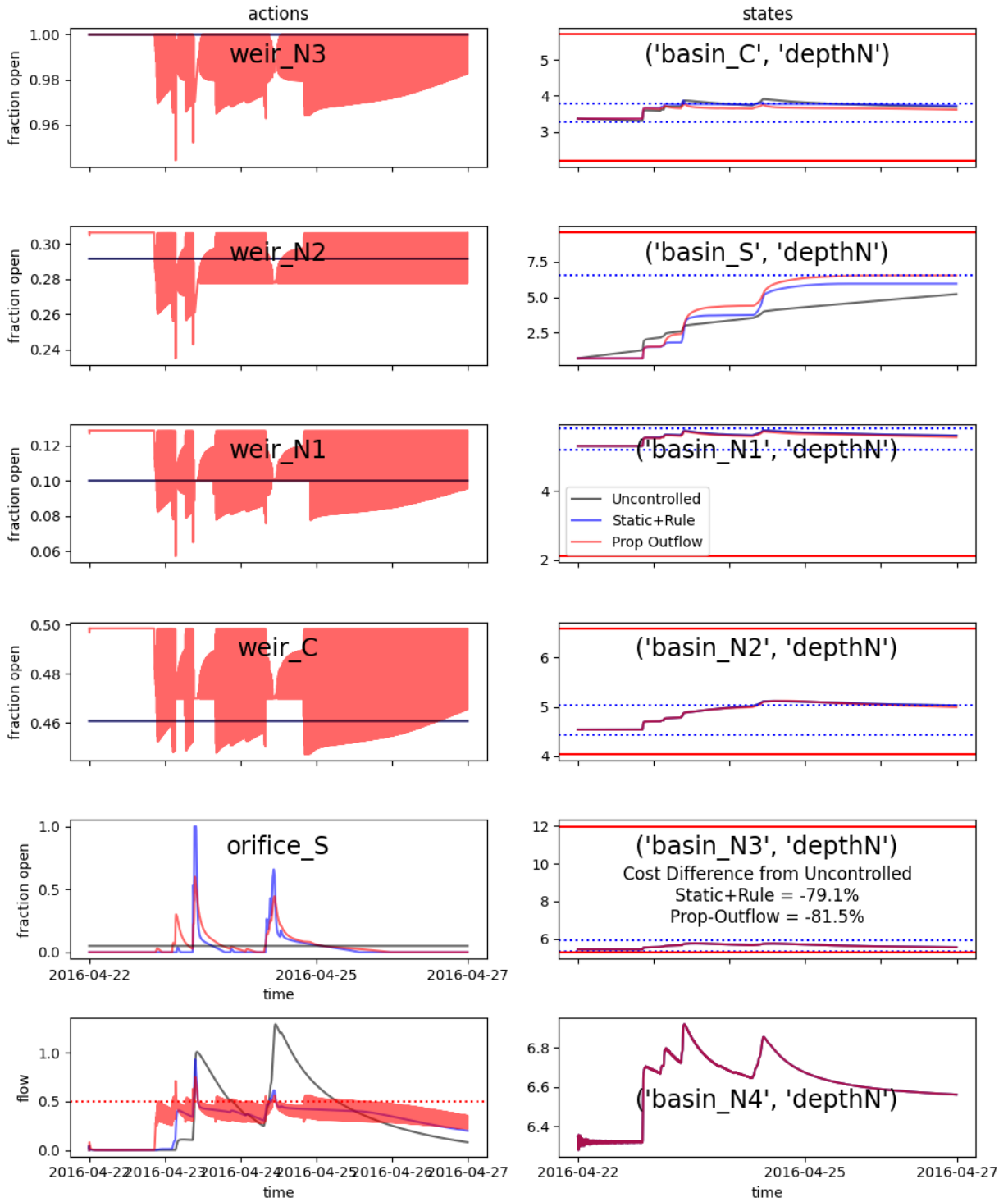


Figure 27: Delta noise level 1

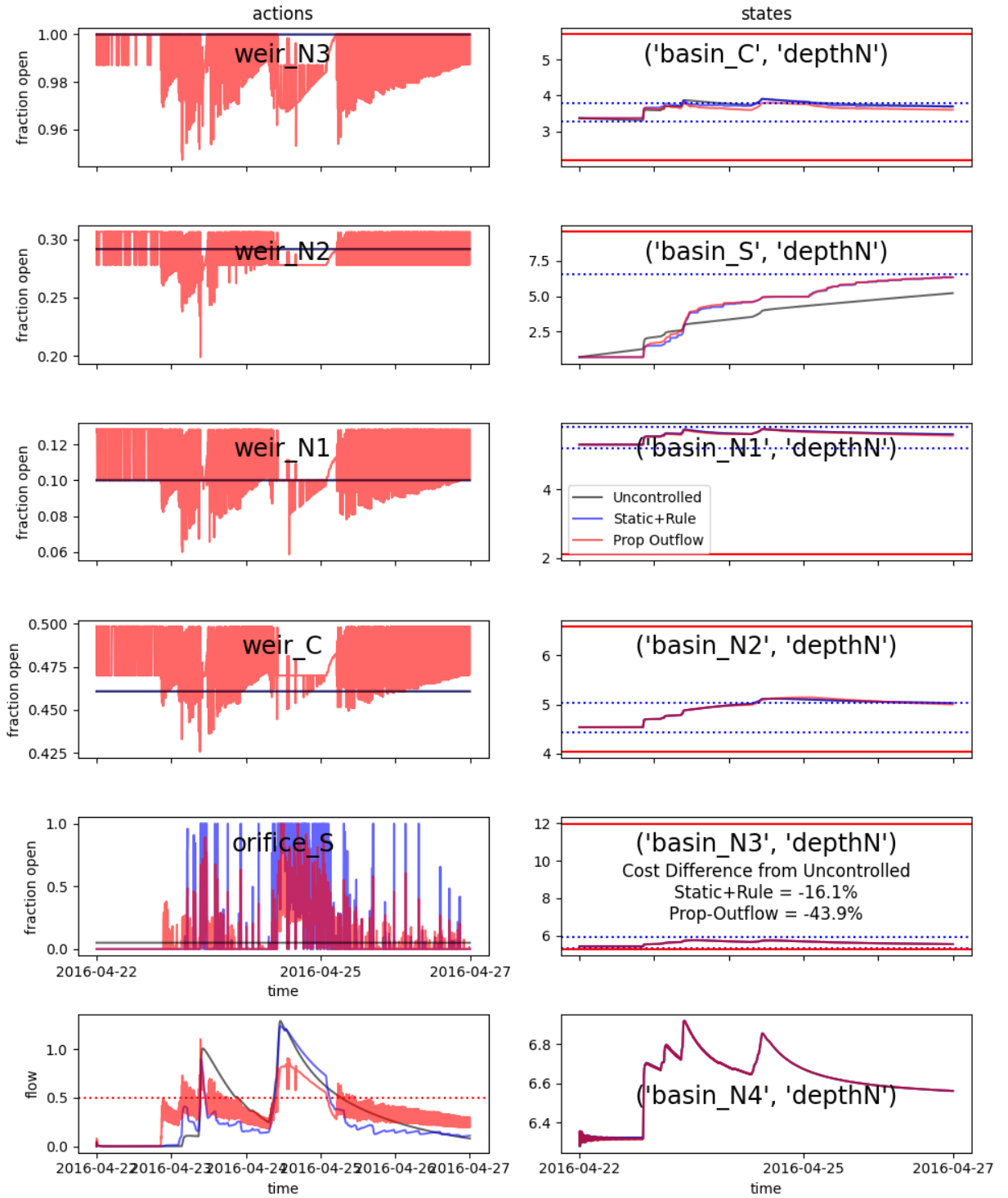


Figure 28: Delta noise level 2

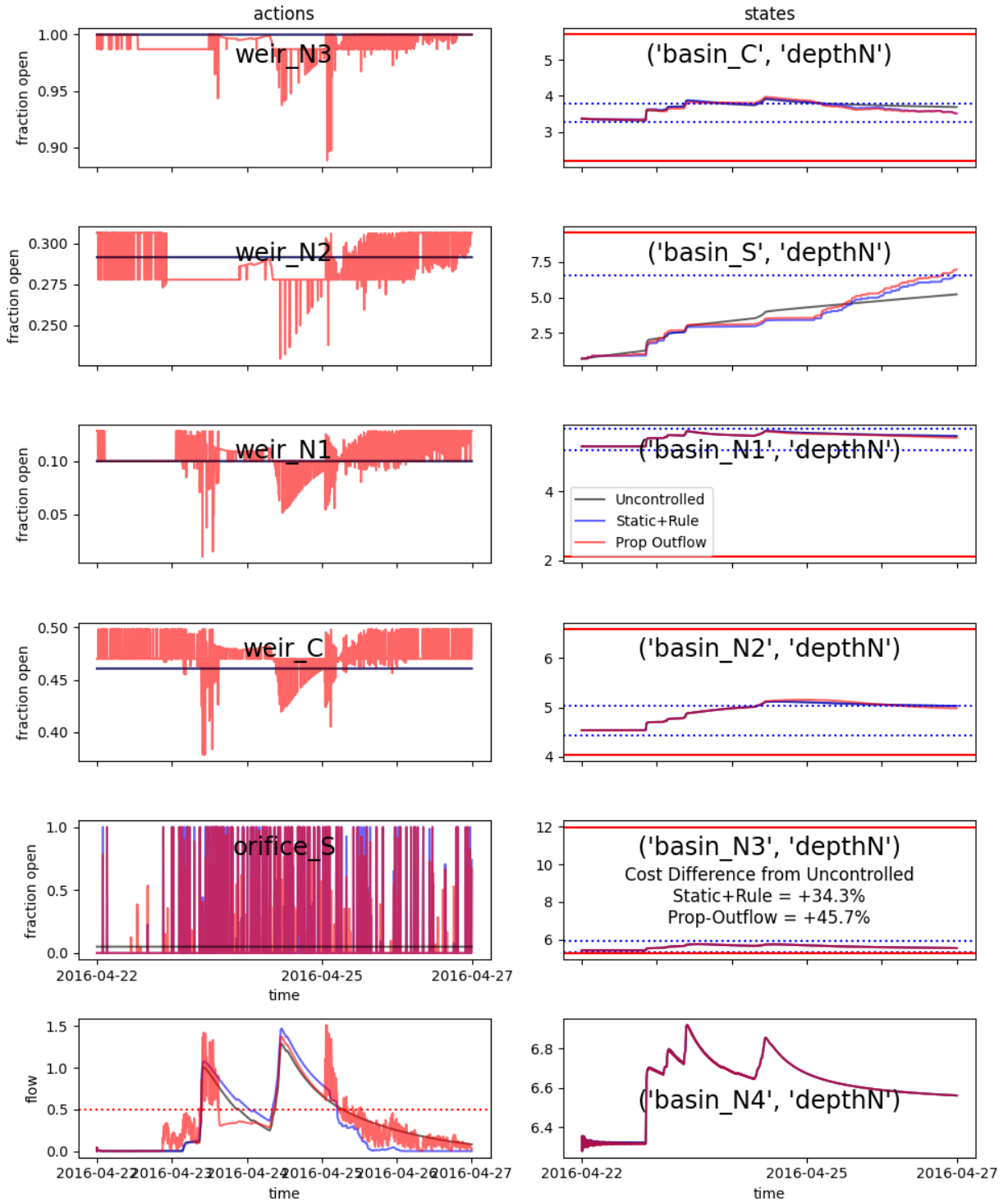


Figure 29: Delta noise level 3

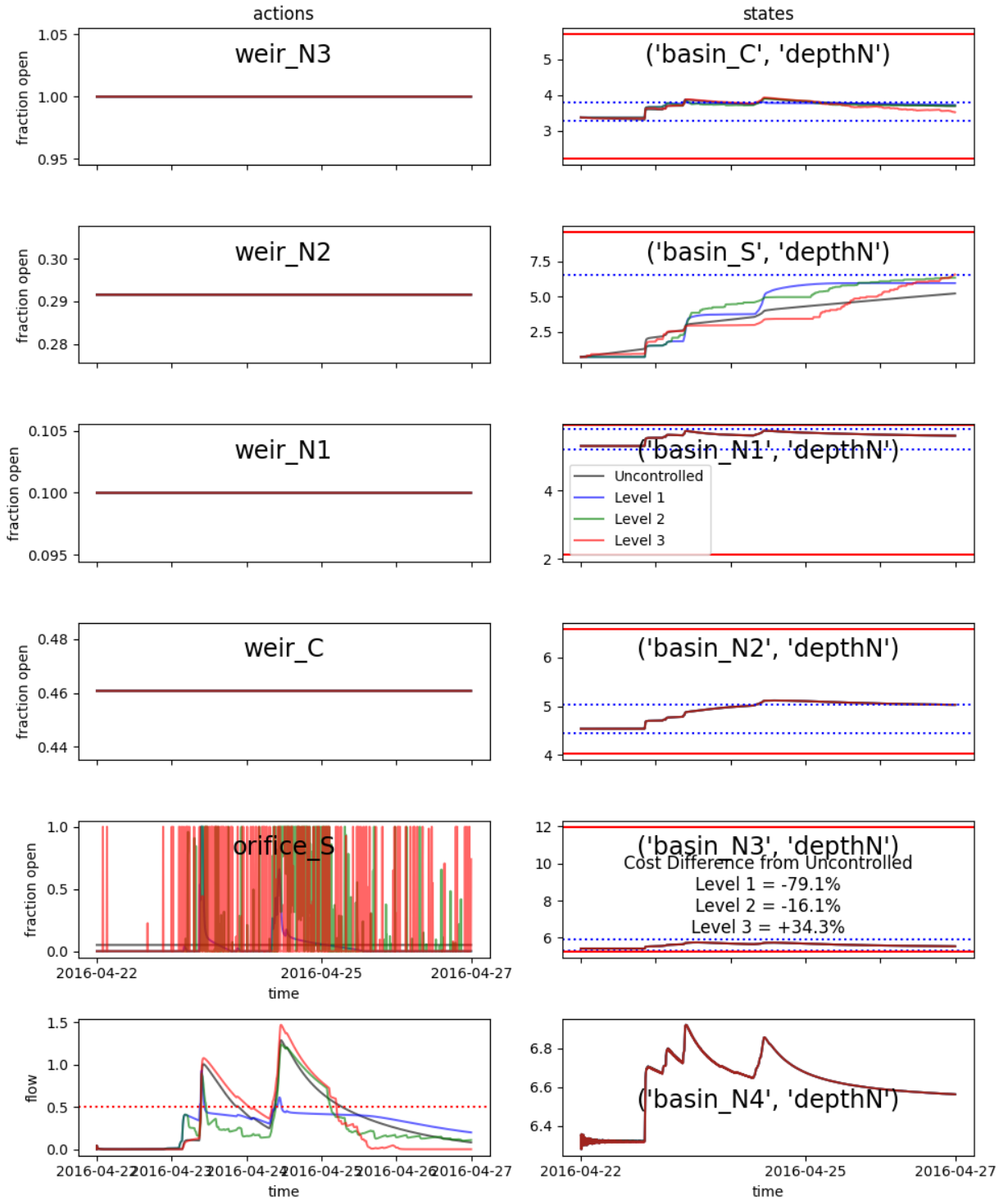


Figure 30: Delta static plus rule based control

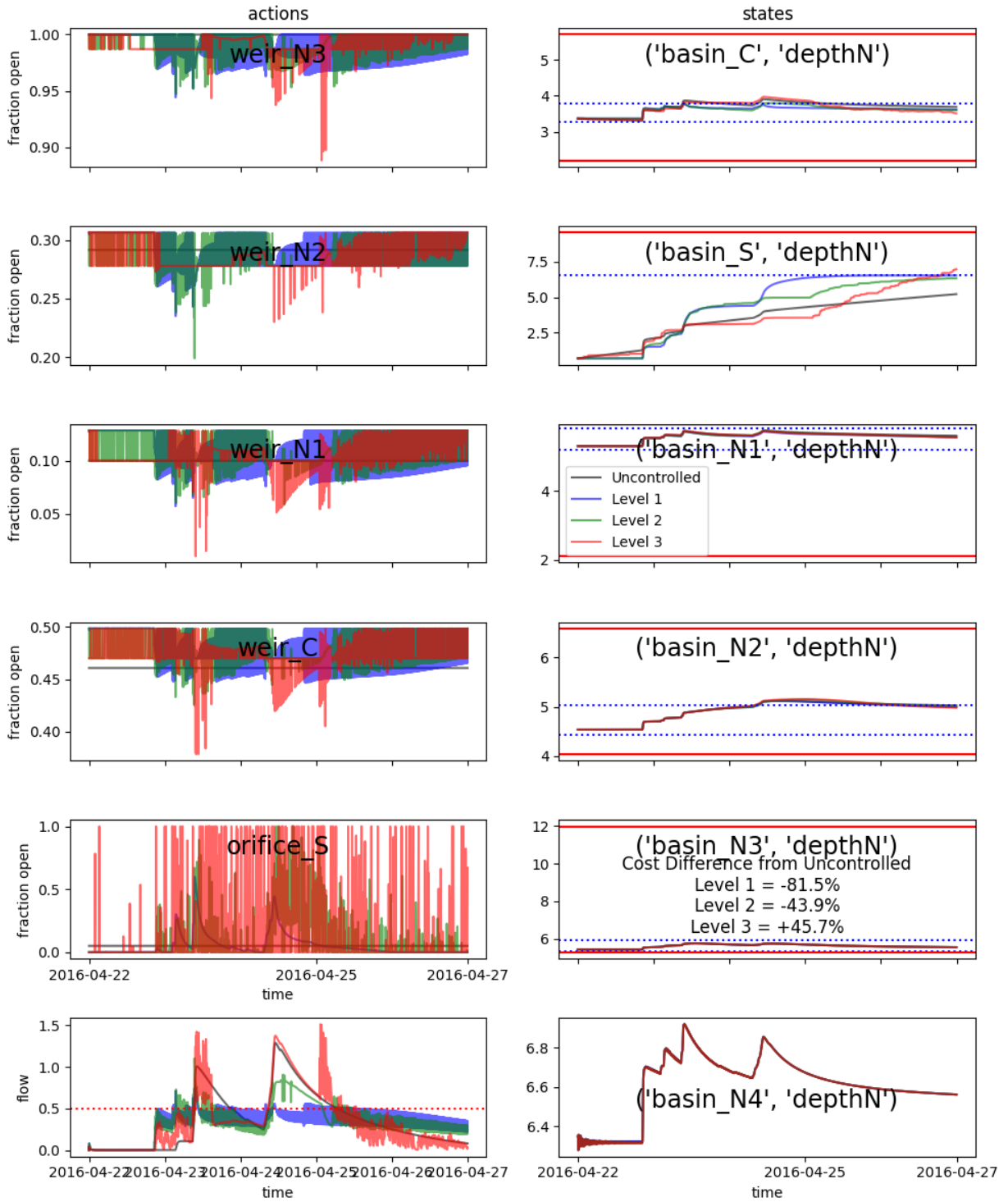


Figure 31: Delta proportional outflow feedback control

Bayesian optimization with unknown constraints for stormwater

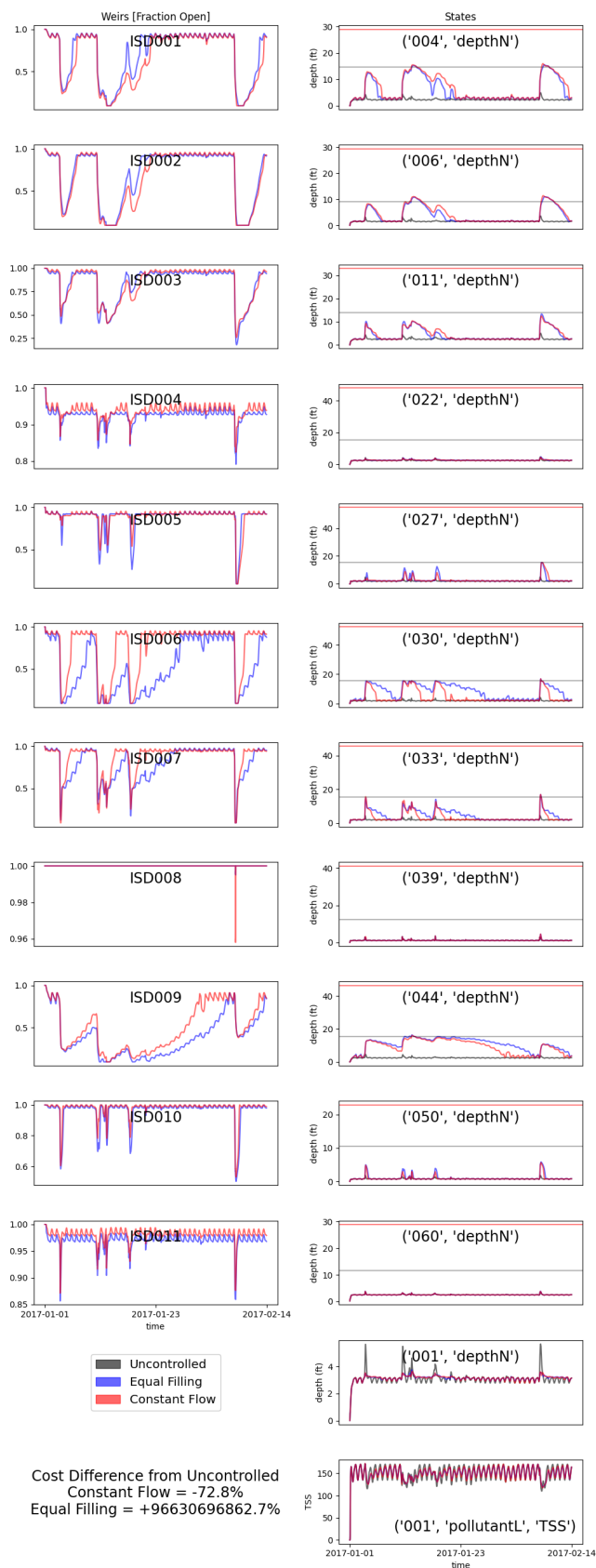


Figure 32: Epsilon noise level 1

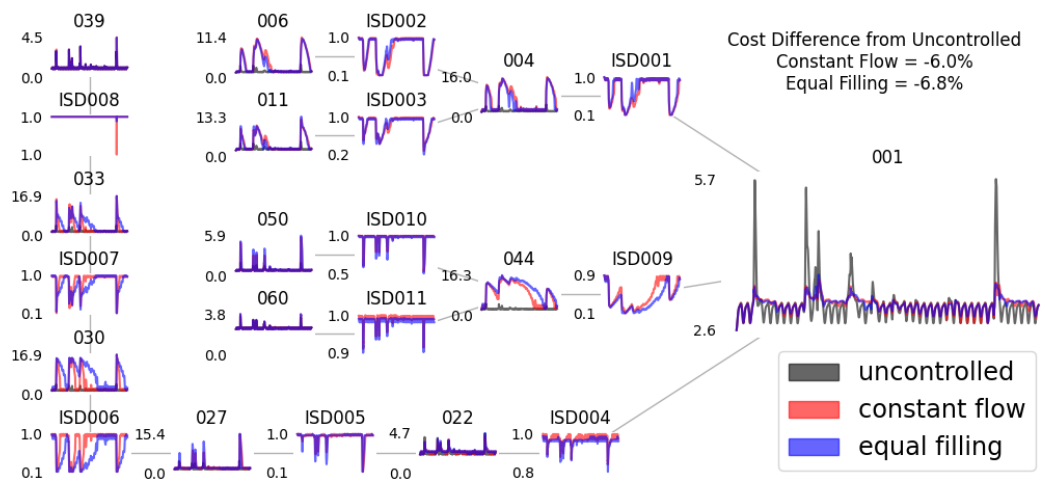


Figure 33: Epsilon noise level 1

Bayesian optimization with unknown constraints for stormwater

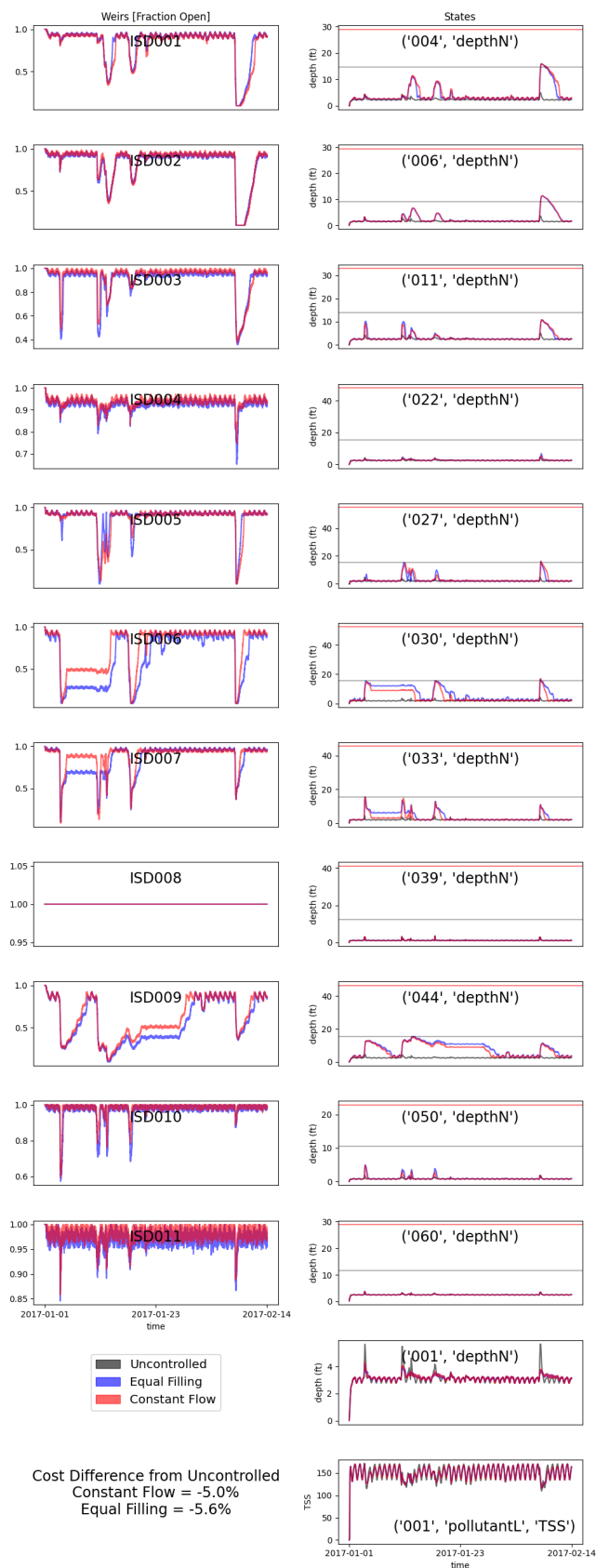


Figure 34: Epsilon noise level 2

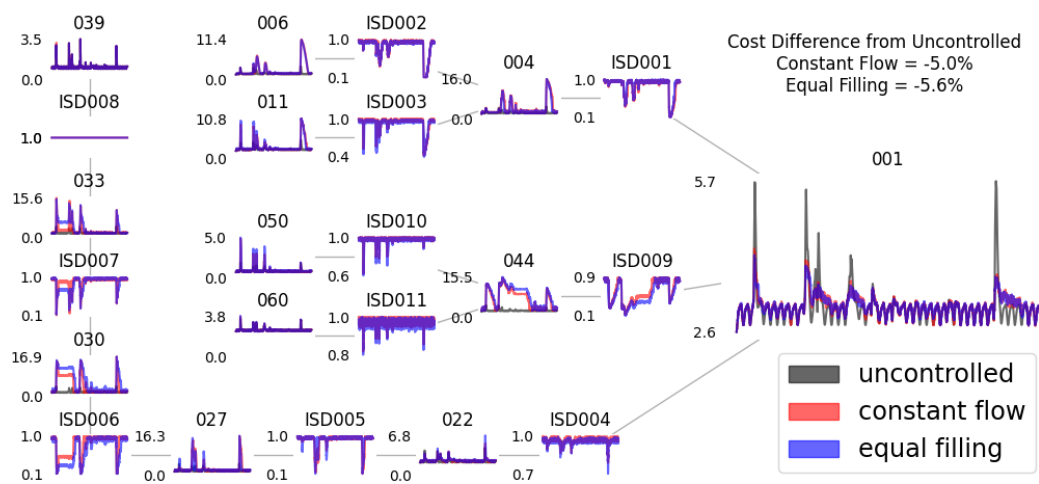


Figure 35: Epsilon noise level 2

Bayesian optimization with unknown constraints for stormwater

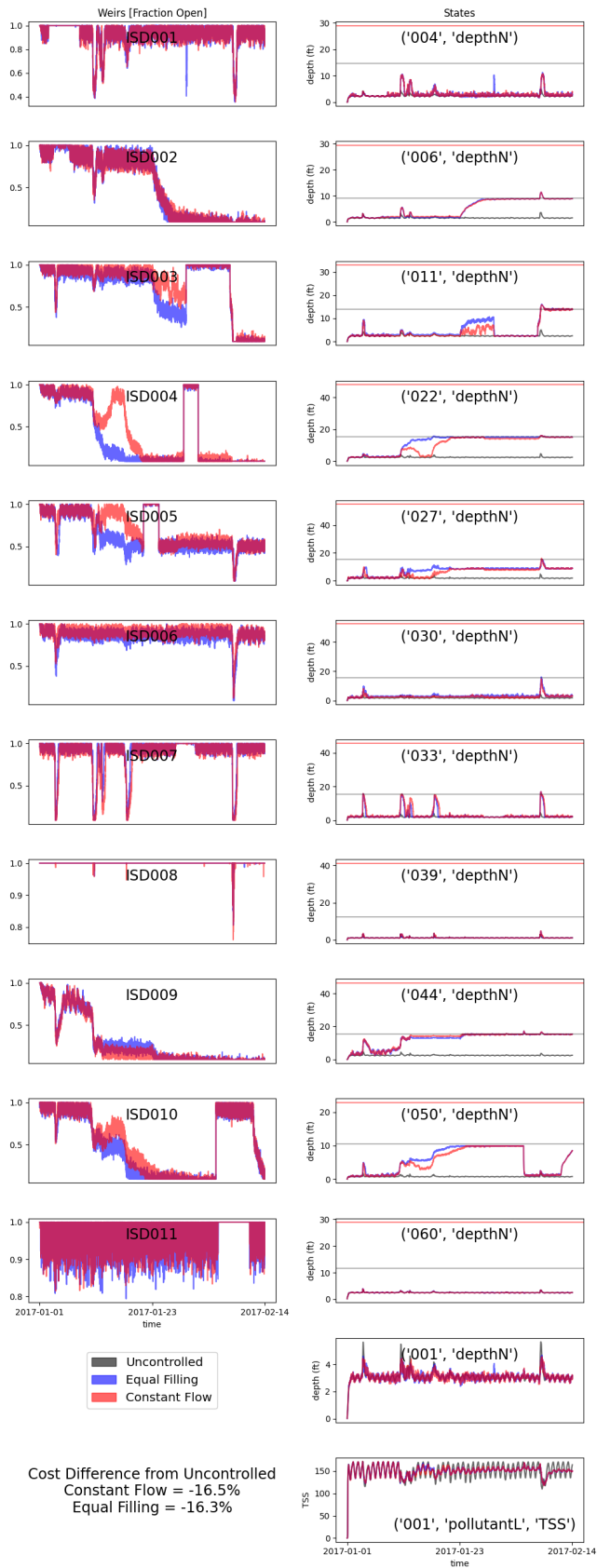


Figure 36: Epsilon noise level 3

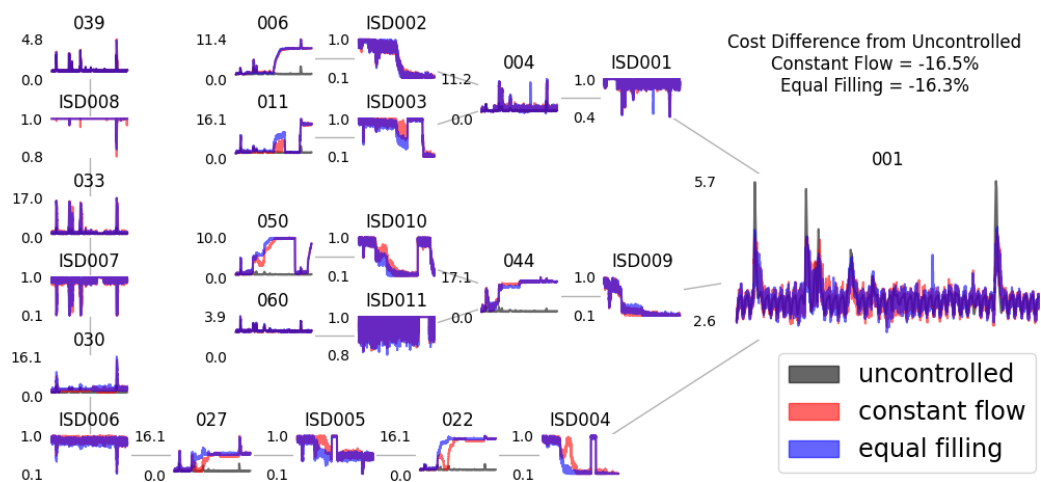


Figure 37: Epsilon noise level 3

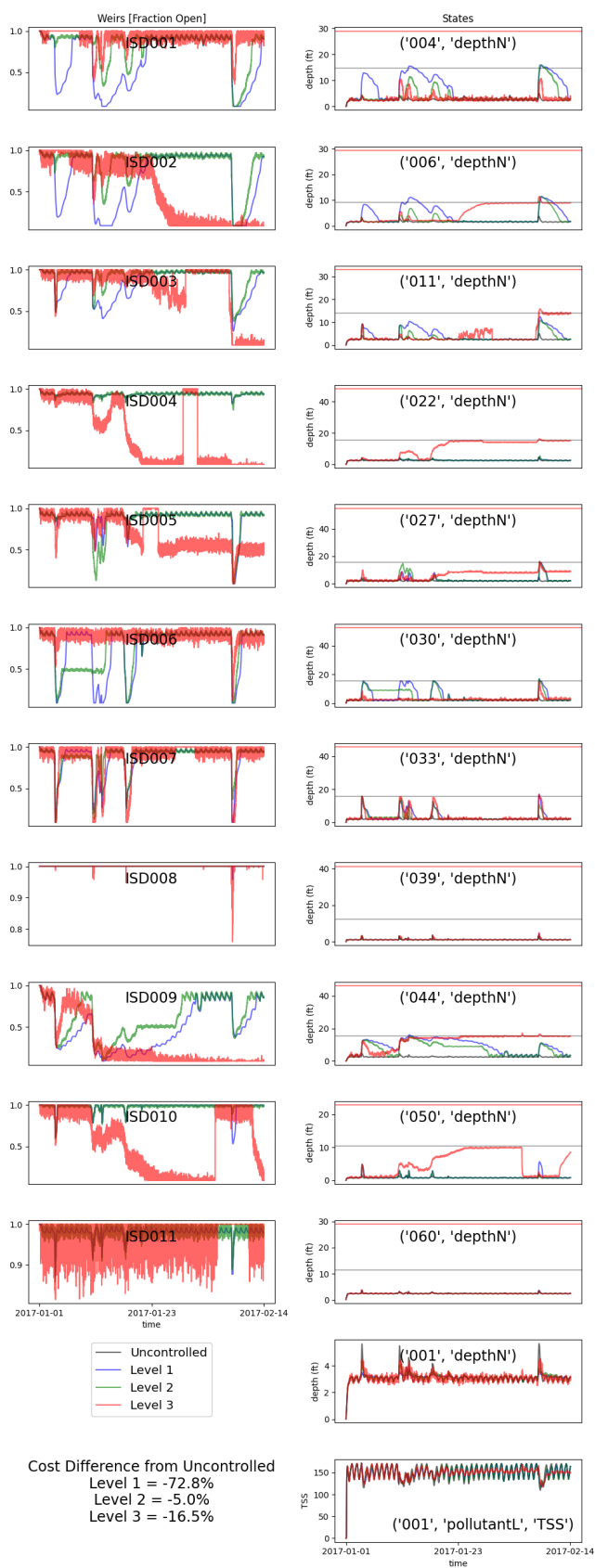


Figure 38: Epsilon constant flow

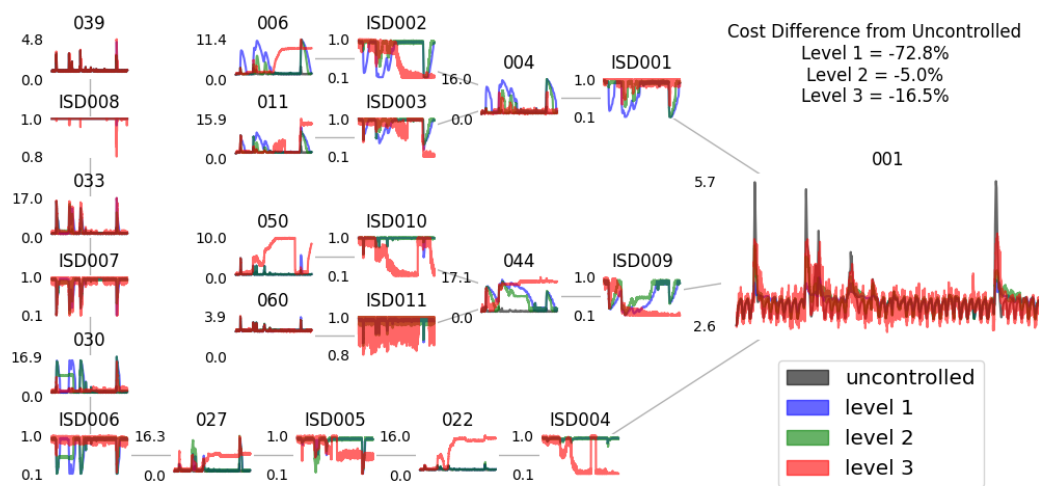


Figure 39: Epsilon constant flow

Bayesian optimization with unknown constraints for stormwater

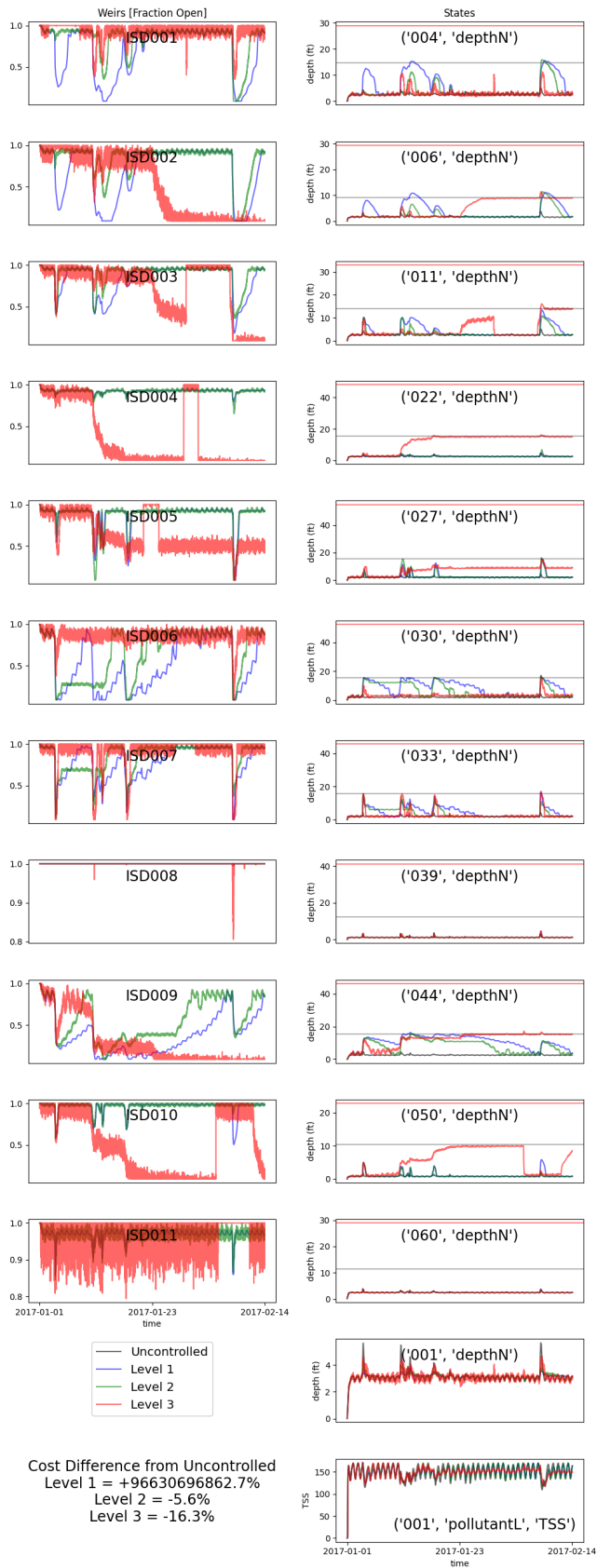


Figure 40: Epsilon equal filling

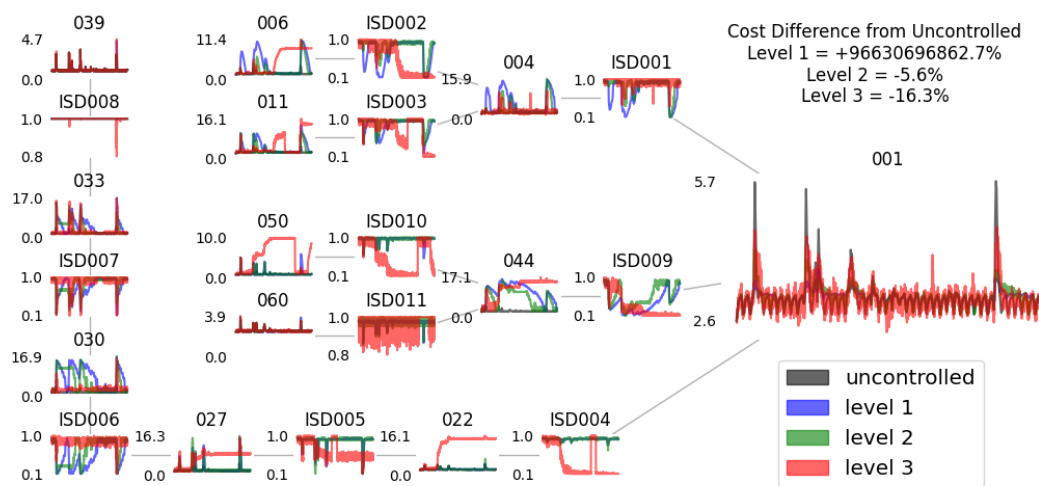


Figure 41: Epsilon equal filling